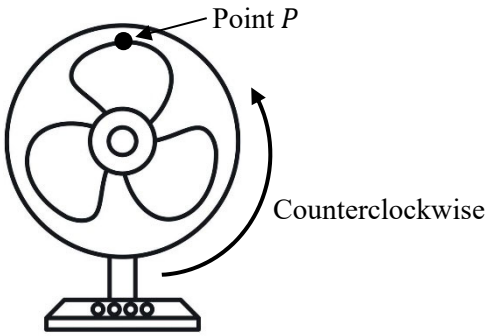
d. Using the above information, create a sinusoidal function model.

e. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate in part d.

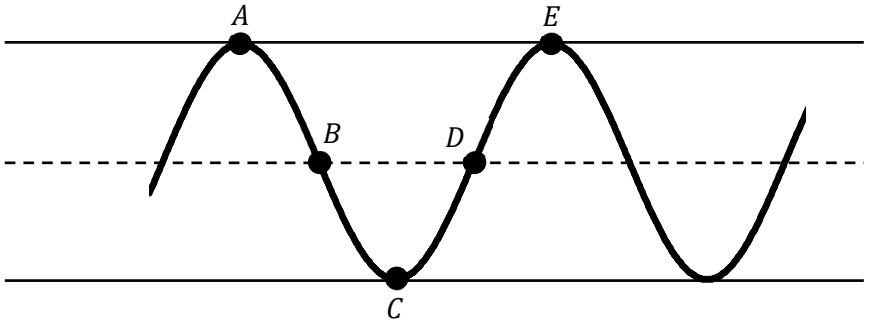
3.7 Sinusoidal Function Context and Data Modeling

3.7 Test Prep

3. **Calculator active.** Mr. Sullivan is tired of not having air conditioning in Germany, so he buys a large fan for his living room. The blades of this fan rotate in a counterclockwise direction and complete 20 rotations every second. Point P is on the tip of one of the fan blades and is located directly above the center of the fan at time $t = 0$ seconds, as indicated in the figure. Point P is 12 inches from the center of the fan. The center of the fan is 30 inches above the floor. As the fan blades rotate at a constant speed, the distance between P and the floor periodically decreases and increases.



The sinusoidal function h models the distance between P and the floor, in inches, as a function of time t, in seconds. The graph of h and its dashed midline for two full cycles is shown. Five points, $A, B, C, D,$ and E, are labeled on the graph. No scale is indicated, and no axes are presented.



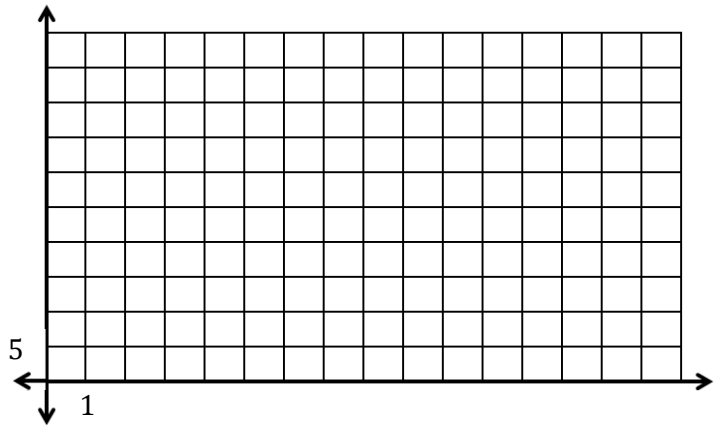
Determine possible coordinates $\big(t, h(t)\big)$ for the five points $A, B, C, D,$ and E

4. A student rides his bicycle to school each day. The pedals are mounted on a bracket whose center is 29 cm above the ground. Each pedal is 16.5 cm from the center of the bracket. Assume that the bicycle is pedaled at 12 cycles per minute. With the starting position of the pedals in a horizontal position at $t = 0$.

   a. Label the graph with time (seconds) on the x-axis. Each line will represent one second.

   b. Label the y-axis as height (cm) from the ground. Each line will represent 5 cm.

   c. Sketch the graph of this sinusoidal function for the first three cycles. Assume the pedal you are measuring goes "up" on the first rotation.





   d. Write the equation of the sine curve with the time given in seconds.

   e. **Calculator active.** When is the pedal 40 cm above the ground for the 3rd time?

   f. **Calculator active.** How high is the pedal after 23 seconds?

---

5. A robotic ant is designed to cross a table in a sinuisoidal pattern, making a wave pattern as it travels from left to right. The table has a length of 1,800 mm and a width of 800 mm. Standing on one side of the table, the values $x = 0$ and $x = 1{,}800$ represent the left and right sides of the table, respectively. The values $y = 0$ and $y = 800$ represent the nearest and furthest sides of the table, respectively.

   The path of the robotic ant begins on the left side of the table, $x = 0$, and completes one period of a sinusoidal function by ending on the right side of the table, $x = 1{,}800$. During its path, the robotic ant reaches its maximum distance from the near side of the table of $y = 750$ before reaching its minimum distance of $y = 150$. If $y = f(x)$ models the path of the robotic ant, which of the following could define $f(x)$?

   (A) $300 \sin\left(\frac{\pi}{900}x\right) + 450$

   (B) $300 \sin(3600\pi x) + 450$

   (C) $600 \sin\left(\frac{\pi}{900}x\right) + 450$

   (D) $600 \sin(3600\pi x) + 450$

6. **Calculator active.** The function f is given by $f(x) = 4\sin(2x) + \cos(6x)$. Using the period of f, how many number of complete cycles of the graph of f in the xy-plane are on the interval $0 \leq x \leq 700$? Round to the nearest whole number.

---

7. When an earthquake hits and creates a tsunami, the water first goes down from its normal level, and then rises an equal distance above its normal level, then returns to its normal level. A tsunami is approaching Trig Island and is modeled by:

$$d(t) = 10\sin\left[\frac{2\pi}{15}(t - 7.5)\right] + 11$$

where d is the water's depth in meters and t is the time in minutes since the earthquake. This model only works for one cycle!



a. Using the equation, what is the maximum depth of the tsunami wave? (No calculator!)

b. Using the equation, what is the normal depth of water at Trig Island? (No calculator!)

c. What is the period of this function? What does this mean in the context of this problem?

d. The Treasure of Trig Island lies 5 meters beneath the normal surface of the water. For how long will the Treasure of Trig Island be exposed? (Remember to look at your answer from part b.)

8. The function g is given by $g(\theta) = \sin(\theta)$. The sinusoidal function h is a phase shift of the function g by positive $\frac{\pi}{6}$ units. Which of the following is true?

(A) Consecutive minima of h occur at $\left(-\frac{5\pi}{6}, g\left(-\frac{5\pi}{6}\right)\right)$ and $\left(\frac{7\pi}{6}, g\left(\frac{7\pi}{6}\right)\right)$ because consecutive minima of g occur at $\left(-\pi, g(-\pi)\right)$ and $\left(\pi, g(\pi)\right)$, and h is the image of g with a horizontal shift of $\frac{\pi}{6}$ units right.

(B) Consecutive minima of h occur at $\left(-\frac{7\pi}{6}, g\left(-\frac{7\pi}{6}\right)\right)$ and $\left(\frac{5\pi}{6}, g\left(\frac{5\pi}{6}\right)\right)$ because consecutive minima of g occur at $\left(-\pi, g(-\pi)\right)$ and $\left(\pi, g(\pi)\right)$, and h is the image of g with a horizontal shift of $\frac{\pi}{6}$ units left.

(C) Consecutive minima of h occur at $\left(-\frac{\pi}{3}, g\left(-\frac{\pi}{3}\right)\right)$ and $\left(\frac{5\pi}{3}, g\left(\frac{5\pi}{3}\right)\right)$ because consecutive minima of g occur at $\left(-\frac{\pi}{2}, g\left(-\frac{\pi}{2}\right)\right)$ and $\left(\frac{3\pi}{2}, g\left(\frac{3\pi}{2}\right)\right)$, and h is the image of g with a horizontal shift of $\frac{\pi}{6}$ units right.

(D) Consecutive minima of h occur at $\left(-\frac{2\pi}{3}, g\left(-\frac{2\pi}{3}\right)\right)$ and $\left(\frac{4\pi}{3}, g\left(\frac{4\pi}{3}\right)\right)$ because consecutive minima of g occur at $\left(-\frac{\pi}{2}, g\left(-\frac{\pi}{2}\right)\right)$ and $\left(\frac{3\pi}{2}, g\left(\frac{3\pi}{2}\right)\right)$, and h is the image of g with a horizontal shift of $\frac{\pi}{6}$ units left.

---

9. The function f is defined by $f(x) = a\sin(b(x + c) + d$. In the xy-plane, the points $(3,6)$ and $(5,16)$ represent a minimum value and a maximum value, respectively, on the graph of f. What are values of a and d?

3.7 Sinusoidal Function Context and Data Modeling

AP Precalculus Name: _____ **CA #1**

1. The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	-60	-44	-27	-9	7	20	10	-8	-16	-40

x	10	11	12	13	14	15	16	17	18	19
$f(x)$	-58	-41	-12	-10	9	18	4	-5	-24	-49

a. Estimate the period and frequency.

b. Estimate the vertical shift (midline).

c. Estimate the amplitude.

d. Using the above information, create a sinusoidal function model.

e. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate in part d.

2. The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	-294	-205	-168	-129	-109	-135	-207	-246	-301	-370

x	10	11	12	13	14	15	16	17	18	19
$f(x)$	-396	-335	-285	-250	-190	-131	-97	-122	-196	-251

a. Estimate the period and frequency.

b. Estimate the vertical shift (midline).

c. Estimate the amplitude.

d. Using the above information, create a sinusoidal function model.

e. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate in part d.

Answers to 3.7 CA #1

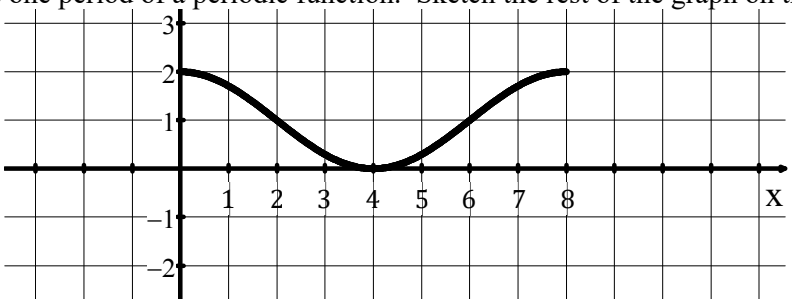
1a. Period: 10 Frequency: $\frac{1}{10}$	1b. down -20	1c. 40	1d. $f(x) = 40\sin\left(\frac{\pi}{5}[x-2.5]\right) - 20$ or $f(x) = 40\cos\left(\frac{\pi}{5}[x-5]\right) - 20$
1e. $f(x) = 33.416\sin(0.637x - 1.657) - 16.322$			
2a. Period: 12 Frequency: $\frac{1}{12}$	2b. down -250	2c. 150	2d. $f(x) = 150\sin\left(\frac{\pi}{6}[x-1]\right) - 250$ or $f(x) = 150\cos\left(\frac{\pi}{6}[x-4]\right) - 250$
2e. $f(x) = 130.386\sin(0.516x - 0.36) - 243.588$			

Name: _____ Date:_____ Period: _____ **3A Review**

Unit 3A Review – Trigonometric and Polar Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 3.1 – 3.2.

1. The graph below shows one period of a periodic function. Sketch the rest of the graph on the given axes.



   a. Is the function above increasing, decreasing or both on the interval $20 < x < 24$.

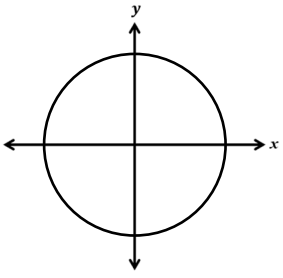
The measurement of an angle in standard position is listed. In which quadrant is the terminal ray?

2. 6.6π

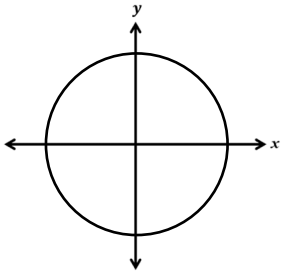
3. -2.3π

Below are various measurements of a circle's radius, an angle within the circle, or the arc subtended by the angle. SKETCH the approximate angle on the axes and find the missing value.

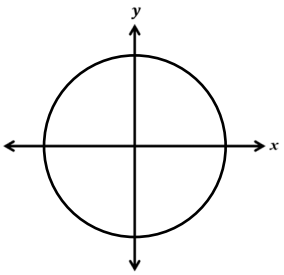
4. Radius is 8.2 and the length of an arc subtended by an angle is 37.63. Find the measure of the angle.



5. Radius is 5.27 and an angle is 0.56π radians. Find the length of the arc subtended by the angle.

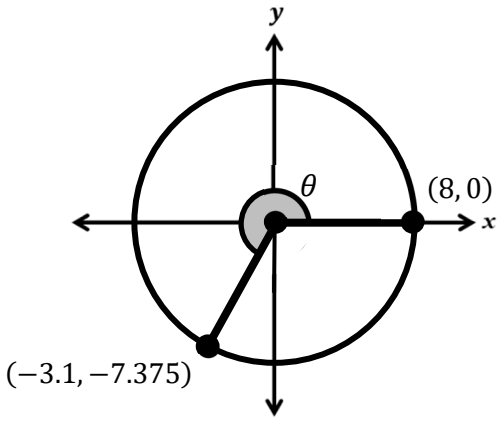


6. An angle is 3.56 radians and the length of an arc subtended by the angle is 5.2. What is the radius of the circle?



Each figure below gives a circle in the xy-plane with center at the origin, and an angle θ in standard position. Find the value of each expression.

7.

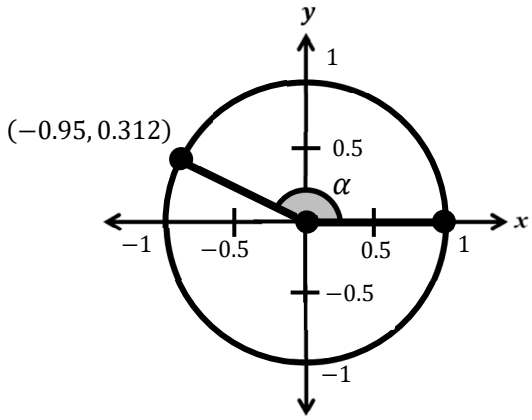


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

8.



a. $\sin\alpha =$

b. $\cos\alpha =$

c. $\tan\alpha =$

Find the value of each expression. Do not look back at the Unit Circle for help!

9. $\cos\pi$	10. $\sin\frac{\pi}{4}$	11. $\cos\frac{2\pi}{3}$	12. $\cos\frac{7\pi}{6}$	13. $\sin\left(-\frac{\pi}{6}\right)$	14. $\cos\left(-\frac{5\pi}{4}\right)$

For each problem, an angle in standard position in the xy-plane is given in radians. A circle is centered at the origin with the given radius. What are the coordinates of the point of intersection of the terminal ray of the angle and the circle?

15. $\theta = \frac{3\pi}{2}, r = 6$

16. $\theta = \frac{5\pi}{3}, r = 10$

In the xy-plane, the terminal ray of angle θ in standard position intersects a circle of radius r at the given point. What are the values of θ and r, with $0 \leq \theta \leq 2\pi$?

17. $(\sqrt{3}, -1)$

18. $(-6, 6\sqrt{3})$

The function f is given by $f(\theta) = \cos\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

19. $\frac{\pi}{2} < \theta < \pi$

20. $\frac{3\pi}{2} < \theta < 2\pi$

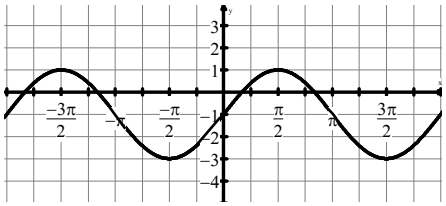
The function f is given by $f(\theta) = \sin\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

21. $0 < \theta < \frac{\pi}{2}$

22. $\pi < \theta < \frac{3\pi}{2}$

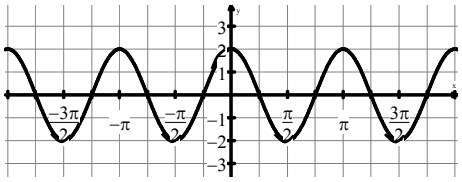
Write the given function for each graph. Use a positive coefficient for all equations.

23. Write a **sine** function.



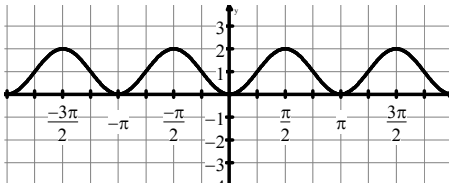
$y =$ _____

24. Write a **cosine** function.



$y =$ _____

25. Write a **sine** function.



$y =$ _____

State the amplitude, period, phase shift, and vertical shift.

26. $y = 3\cos(5x - \pi)$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

Phase shift: _____

27. $y = \frac{1}{2}\sin(4x) - 2$

Amp: _____ Period: _____

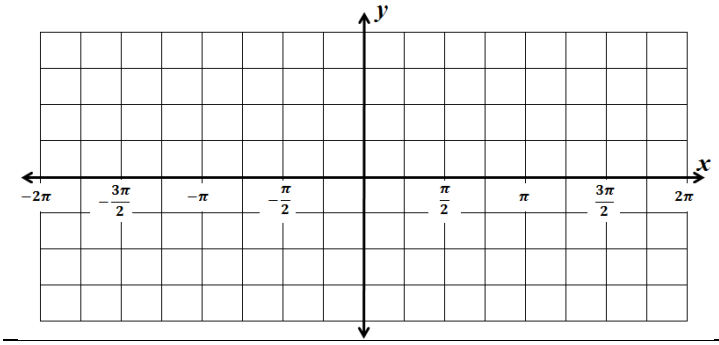
Midline: _____ Freq: _____

Max value: _____ Min value: _____

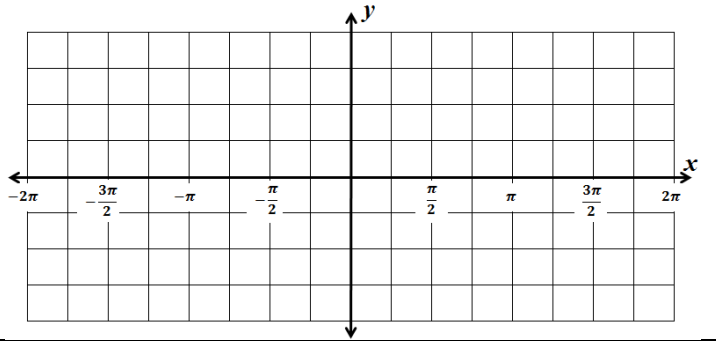
Phase shift: _____

Graph each function on the given set of axes.

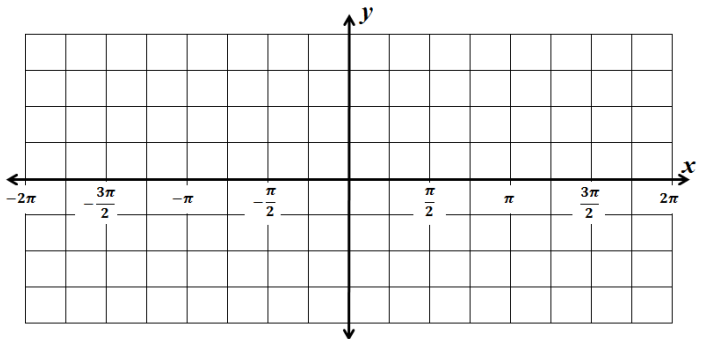
28. $y = 2\cos x$



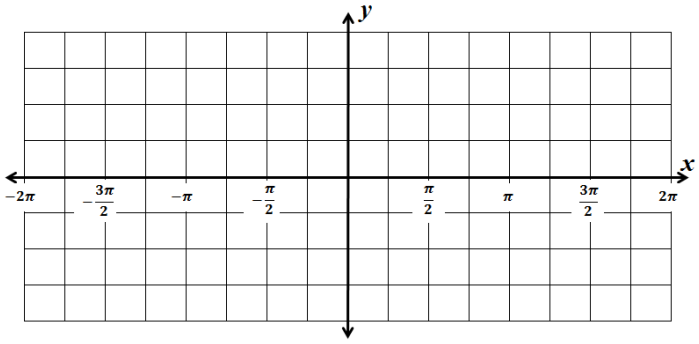
29. $y = 3\sin 2x - 1$



30. $y = -3\sin\left(\frac{x}{2} + \pi\right)$



31. $y = 2\cos(2x - \pi) - 1$



32. Suppose you are riding a Ferris wheel. After everyone is loaded, the wheel starts to turn, and the ride lasts for 105 seconds. Your height h (in feet) above the ground at any time t (in seconds) can be modeled by the equation $h(t) = 50\sin\left[\frac{\pi}{10}(t-4)\right] + 65$. You do not need a calculator for any question except the last two.

a. What is the period?

b. What does the period represent?

c. What is the frequency?

d. What does the frequency represent?

e. What is your maximum height?

f. What is your minimum height?

g. How many circles will the Ferris Wheel make during the ride?

h. **Calculator active.** How high are you when the ride begins?

i. **Calculator acctive.** What is your height when the ride stops?

33. The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	16	2	-10	-1	18	21	40	52	37	29

x	10	11	12	13	14	15	16	17	18	19
$f(x)$	12	-2	-11	4	17	22	38	49	39	26

a. Estimate the period and frequency.

b. Estimate the vertical shift (midline).

c. Estimate the amplitude.

d. Using the above information, create a sinusoidal function model.

e. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate in part d.

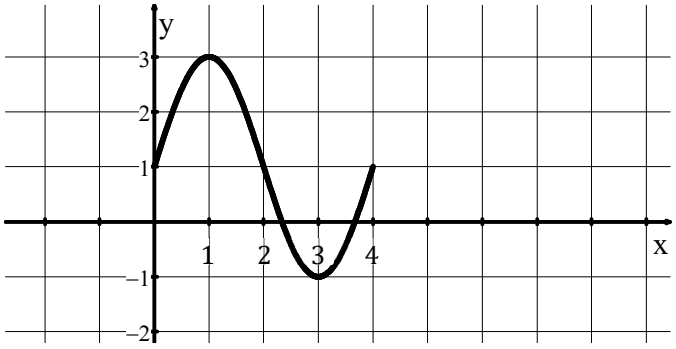
Name: _____ Date:_____

Corrective Assignment

Unit 3A CA – Trigonometric and Polar Functions

1.

a. The graph below shows one period of a periodic function. Sketch the rest of the graph on the given axes.

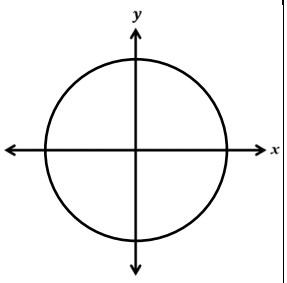


b. Is the function concave up, concave down, or both on the interval $42 < x < 44$?

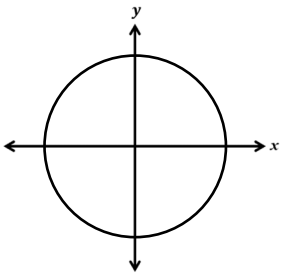
2. An angle in standard position with a measure of -7.2π would have a terminal ray in which quadrant?

Below are various measurements of a circle's radius, an angle within the circle, or the arc subtended by the angle. SKETCH the approximate angle on the axes and find the missing value.

3. Radius is 9.4 and the length of an arc subtended by an angle is 42.7. Find the measure of the angle.



4. Radius is 1.26 and an angle is 4.05 radians. Find the length of the arc subtended by the angle.

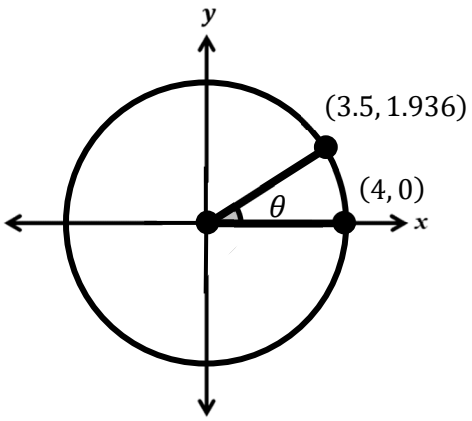


Find the value of each expression. Try not to look back at the Unit Circle for help.

5. $\sin(-\pi)$	6. $\cos\frac{\pi}{6}$	7. $\sin\frac{5\pi}{3}$	8. $\cos\frac{11\pi}{6}$	9. $\cos\left(-\frac{5\pi}{6}\right)$	10. $\sin\left(-\frac{3\pi}{4}\right)$

Each figure below gives a circle in the xy-plane with center at the origin, and an angle θ in standard position. Find the value of each expression.

11.

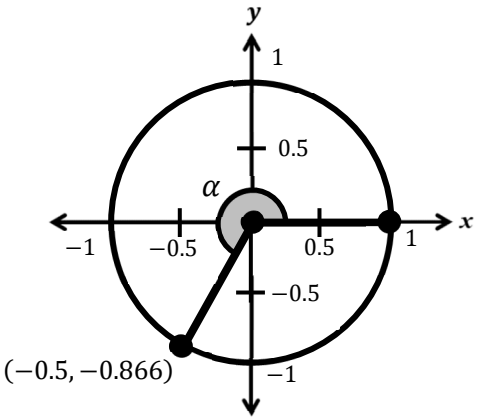


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

12.



a. $\sin\alpha =$

b. $\cos\alpha =$

c. $\tan\alpha =$

For each problem, an angle in standard position in the xy-plane is given in radians. A circle is centered at the origin with the given radius. What are the coordinates of the point of intersection of the terminal ray of the angle and the circle?

13. $\theta = \frac{\pi}{2}, r = 8$

14. $\theta = \frac{3\pi}{4}, r = 22$

In the xy-plane, the terminal ray of angle θ in standard position intersects a circle of radius r at the given point. What are the values of θ and r?

15. $(-3, -3\sqrt{3})$

16. $(-7, 0)$

The function f is given by $f(\theta) = \cos\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

17. $\frac{\pi}{2} < \theta < \pi$

18. $\pi < \theta < \frac{3\pi}{2}$

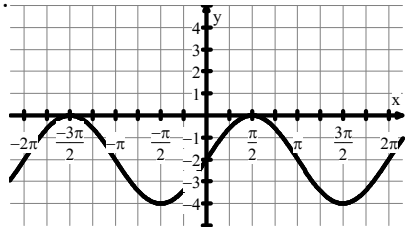
The function f is given by $f(\theta) = \sin\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

19. $0 < \theta < \frac{\pi}{2}$

20. $\frac{3\pi}{2} < \theta < 2\pi$

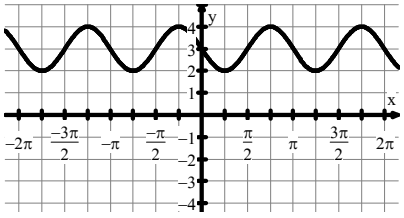
Write a <u>SINE</u> function for each graph. If needed use a phase shift, not a negative coefficient.

21.



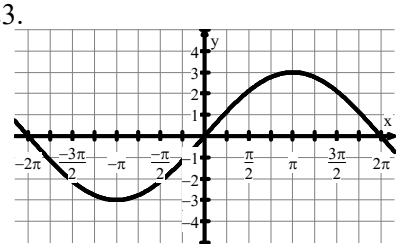
$y =$

22.



$y =$

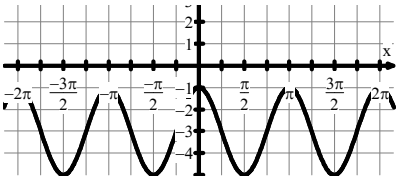
23.



$y =$

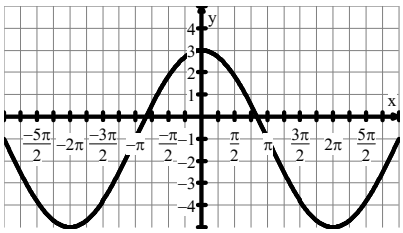
Write a <u>COSINE</u> function for each graph. If needed use a phase shift, not a negative coefficient.

24.



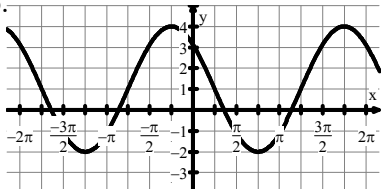
$y =$

25.



$y =$

26.



$y =$

27. Write the equation of a **<u>sine</u>** curve with the following transformations:

- Move down 5
- Move right $\frac{\pi}{7}$

$y =$

28. Write the equation of a **<u>cosine</u>** curve with the following transformations:

- One full period occurs 5 times between 0 and 2π.
- Vertical shift up 1.

$y =$

Identify the listed information for each trig function.

29. $y = 2\sin(\pi x - \pi) - 2$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____Min value: _____

Phase shift: _____

30. $y = 4 - 3\cos 5x$

Amp: _____ Period: _____

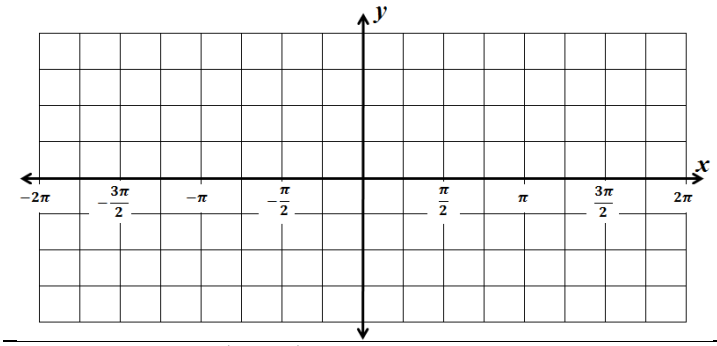
Midline: _____ Freq: _____

Max value: _____Min value: _____

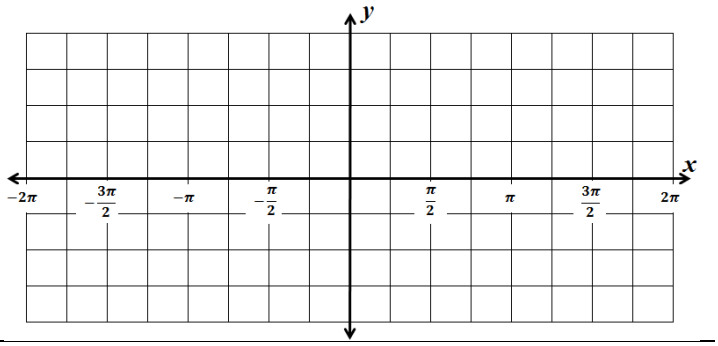
Phase shift: _____

Graph the function.

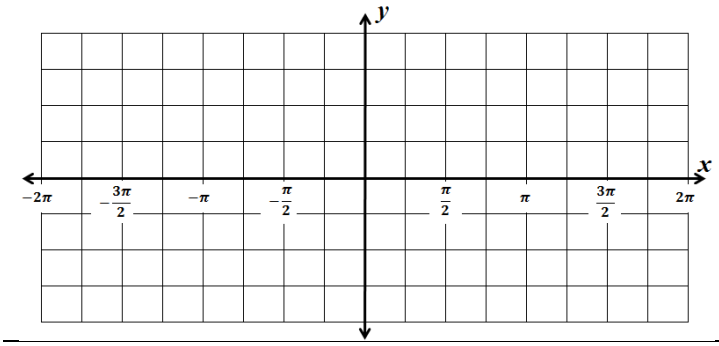
31. $y = 2\sin x + 2$



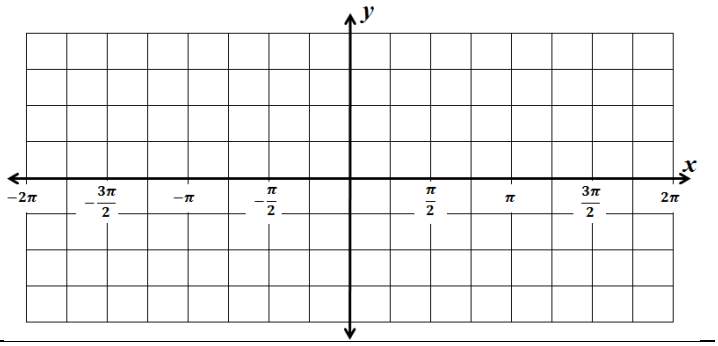
32. $y = -3\cos x$



33. $y = -2\sin\left(x - \frac{\pi}{4}\right)$



34. $y = 2\cos(2x + \pi) - 1$



35. After getting knocked off his boogie board, Mr. Kelly watches it float up and down on the ocean waves. The board moves 6.8 feet from its low point to its high point, returning to its low point every 7 seconds.

a. Write an equation that gives the board's vertical position y at time t if the board is at its lowest point when $t = 0$.

b. Explain why you chose $y = a\sin(bt)$ or $y = a\cos(bt)$ for part a.

36. Suppose you are riding a Ferris wheel. After everyone is loaded, the wheel starts to turn and the ride lasts for 150 seconds. Your height h (in feet) above the ground at any time t (in seconds) can be modeled by the equation

$$h(t) = 55\sin\left[\frac{\pi}{30}(t-10)\right] + 63.$$

a. What is the period?

b. What does the period represent in this scenario?

c. What is the frequency?

d. What does the frequency represent in this scenario?

e. What is your maximum height?

f. What is your minimum height?

g. How many circles will the Ferris Wheel make during the ride?

h. **Calculator active.** How high are you when the ride begins? (Use radians.)

i. **Calculator active.** What is your height when the ride stops?

37. The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	-49	250	-50	-349	-51	249	-52	-353	-51	252

x	10	11	12	13	14	15	16	17	18	19
$f(x)$	-48	-347	-51	249	-50	-351	-49	250	-53	-346

a. Estimate the period and frequency.

b. Estimate the vertical shift (midline).

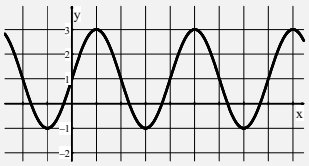
c. Estimate the amplitude.

d. Using the above information, create a sinusoidal function model.

e. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate in part d.

Answers to Unit 3A Corrective Assignment

1a.



1b. Concave up.

2. Quadrant II

3. 4.54 radians

4. 5.103

5. 0

6. $\frac{\sqrt{3}}{2}$

7. $-\frac{\sqrt{3}}{2}$

8. $\frac{\sqrt{3}}{2}$

9. $-\frac{\sqrt{3}}{2}$

10. $-\frac{\sqrt{2}}{2}$

11a. 0.484
11b. 0.875
11c. 0.553

12a. -0.866
12b. -0.5
12c. 1.732

13. $(0, 8)$

14. $\left(-11\sqrt{2}, 11\sqrt{2}\right)$

15. $\theta = \frac{4\pi}{3}, r = 6$

16. $\theta = \pi, r = 7$

17. Concave up
Decreasing

18. Concave up
Increasing

19. Concave down
Increasing

20. Concave up
Increasing

21. $y = 2\sin(x) - 2$

22. $y = \sin(2x - \pi) + 3$

23. $y = 3\sin\left(\frac{1}{2}x\right)$

24. $y = 2\cos(2x) - 3$

25. $y = 4\cos\left(\frac{1}{2}x\right) - 1$

26. $y = 3\cos\left(x + \frac{\pi}{4}\right) + 1$

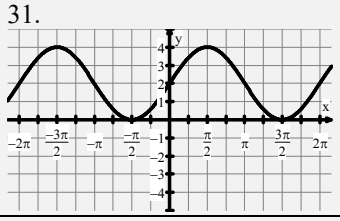
27. $y = \sin\left(x - \frac{\pi}{7}\right) - 5$

28. $y = \cos(5x) + 1$

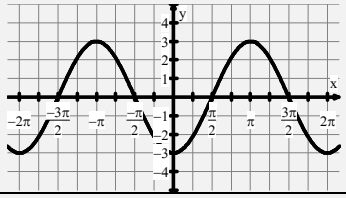
29. Amp: 2 Period: 2
Mid: $y = -2$ Freq: $\frac{1}{2}$
Max: 0 Min: -4
P.S.: right 1

30. Amp: 3 Period: $\frac{2\pi}{5}$
Mid: $y = 4$ Freq: $\frac{5}{2\pi}$
Max: 7 Min: 1
P.S.: NONE

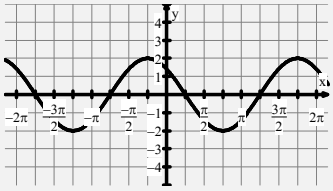
31.



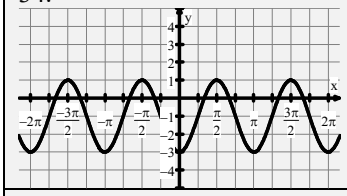
32.



33.



34.



35a. $y = -3.4\cos\left(\frac{2\pi}{7}t\right)$

35b. Because the wave started at its lowest point.

36a. 60
36b. It takes 60 seconds for the Ferris Wheel to complete one circle.
36c. $\frac{1}{60}$
36d. Every second, the Ferris Wheel rotates $\frac{1}{60}$ of a circle.
36e. 118 ft.
36f. 8 ft.
36g. 2.5 circles
36h. 15.369 ft.
36i. 110.631 ft.

37a. Period: 4
Frequency: $\frac{1}{4}$
37b. $y = -50$
37c. 300
37d. $f(x) = 300\sin\left(\frac{\pi}{2}x\right) - 50$ or $f(x) = 300\cos\left(\frac{\pi}{2}[x - 1]\right) - 50$
37e. $f(x) = 299.6\sin(1.571x - 0.001) - 49.969$

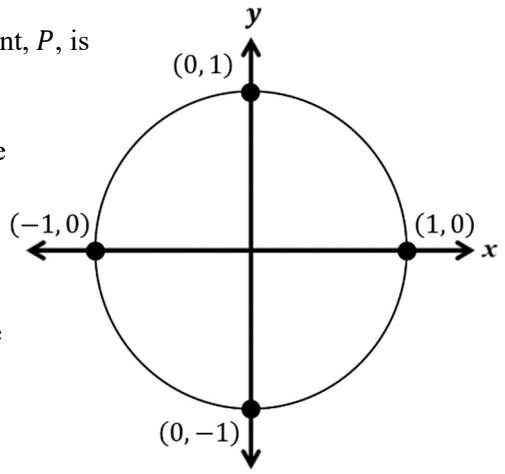
AP Precalc

Write your questions and thoughts here!

3.8 The Tangent Function

3.8 Notes

Take a unit circle and form an angle in standard position. The point, P, is the intersection of the terminal ray and the circle.



The tangent function, $f(\theta) = \tan\theta$, gives the _____ of the terminal ray.

The slope of the terminal ray can also be described by the ratio of the change in y-values to the change in x-values between any two points on the ray. The tangent function is also the ratio of the sine function to the cosine function. Therefore

$$\tan\theta =$$

as long as $\cos\theta \neq 0$.

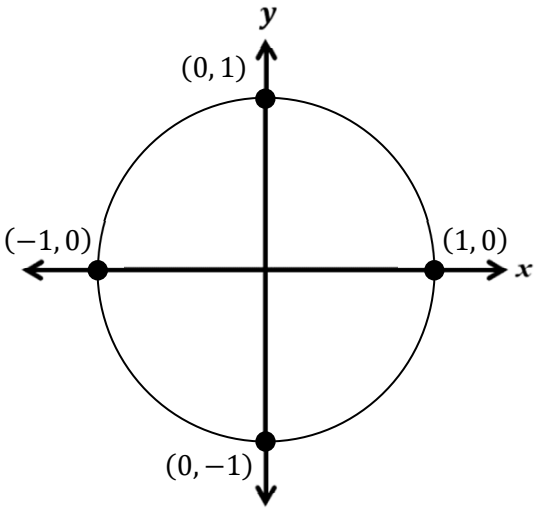
1. In the xy-plane, an angle θ, in standard position, has a measure of $\theta = \frac{\pi}{3}$. What is the slope of the terminal ray of the angle?

Finding the tangent values is the same as finding the slope of the terminal ray of the angle.

Evaluate.		
2. $\tan\frac{\pi}{4}$	3. $\tan\frac{5\pi}{6}$	4. $\tan\frac{3\pi}{2}$

Think about the slope values of the terminal ray as it moves around the unit circle.

- Slope starts as _____.
- Gets larger until it approaches _____.
- At $\theta = \frac{\pi}{2}$, the slope is _____.
- Then the slope is very negative but starts to grow towards zero.
- Once we reach an angle of ___, we are back to a slope of zero again.
- As soon as we pass the angle π, the slope will be the same as it was in _____.
- Every _____ revolution of the circle, the tangent function repeats.

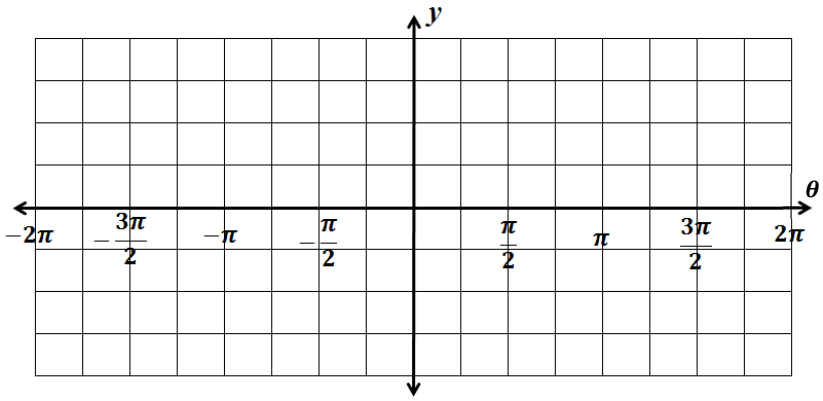


The period of $\tan\theta$ is ____.

Write your questions and thoughts here!

Each time that $\cos\theta = 0$, $\tan\theta$ is undefined. What angle on the unit circle does this occur?

For the graph of $f(\theta) = \tan\theta$, this is represented by vertical asymptotes.



Vertical Asymptotes

For the graph of $f(\theta) = \tan\theta$, a vertical asymptote appears at every $\theta = \frac{\pi}{2} + k\pi$, for integer values of k.

For the graph of $\tan(b\theta)$, the period is

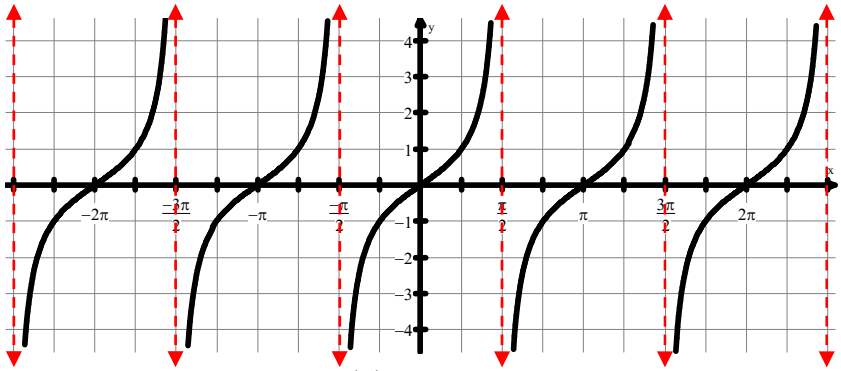
A vertical asymptote appears at every $\theta = \frac{\pi}{\quad} + k\frac{\pi}{\quad}$, for integer values of k.

Write an equation that represents all asymptotes of the graph of f in the xy-plane.

5. $f(\theta) = \tan(2\theta)$

6. $f(\theta) = \tan\left(\frac{2\theta}{3}\right)$

Characteristics of the tangent graph



$f(\theta) = \tan\theta$

- The tangent function increases.
- Its graph changes from concave down to concave up between each set of asymptotes.
- The point where it changes concavity is called an inflection point.

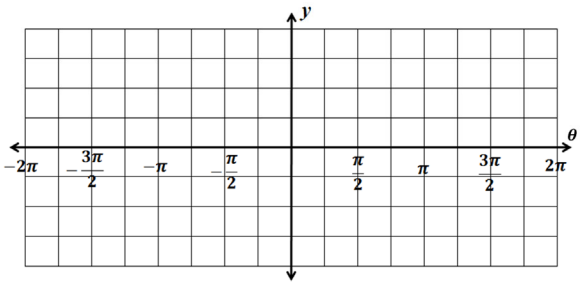
Write your questions and thoughts here!

The graph of $g(\theta) = a\tan\big(b(\theta + c)\big) + d$ is transformation of the graph of $f(\theta) = \tan\theta$ in the following ways:

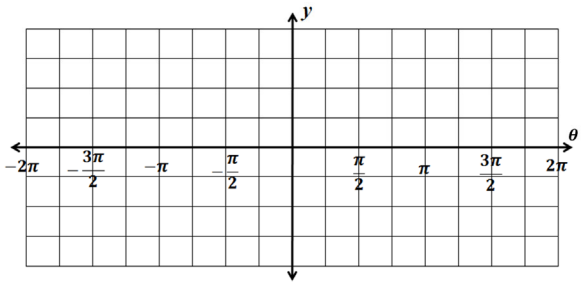
- **The constant a** creates a _____ by a factor of $|a|$. If $a < 0$, there is a reflection over the _____.
- **The constant b** creates a _____ and changes the period by a factor of $\left|\frac{1}{b}\right|$. If $b < 0$, there is a reflection over the _____.
- **The constant c** creates a _____ (phase shift) by $-c$ units.
- **The constant d** creates a _____ by d units.

Graph each function.

7. $y = \frac{1}{2}\tan\left(2\theta - \frac{\pi}{2}\right) + 1$



8. $y = -3\tan\left(\frac{\theta}{2} + \frac{\pi}{2}\right) - 2$



3.8 The Tangent Function

AP Precalculus

3.8 Practice

Write an equation that represents all asymptotes of the graph of f in the xy-plane.

1. $f(\theta) = \tan(3\theta)$

2. $f(\theta) = \tan(6\theta)$

3. $f(\theta) = \tan\left(\frac{\theta}{5}\right)$

In the xy-plane, the angle θ is in standard position. What is the slope of the terminal ray of the angle?

4. $\theta = \frac{\pi}{6}$

5. $\theta = \frac{3\pi}{4}$

6. $\theta = \frac{5\pi}{3}$

Evaluate.

7. $\tan\frac{\pi}{2}$

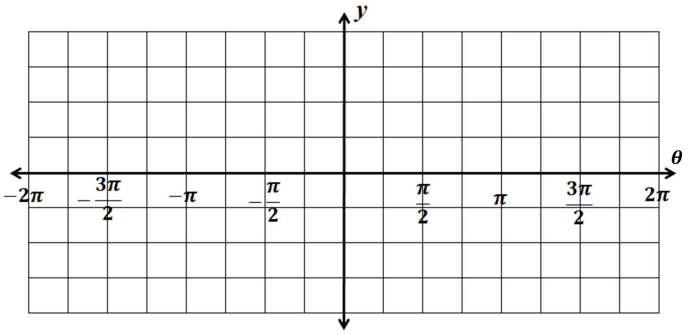
8. $\tan\frac{4\pi}{3}$

9. $\tan\frac{5\pi}{4}$

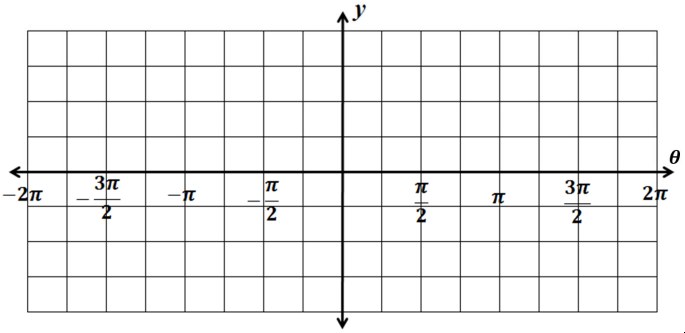
10. $\tan\frac{11\pi}{6}$

Graph each trig function.

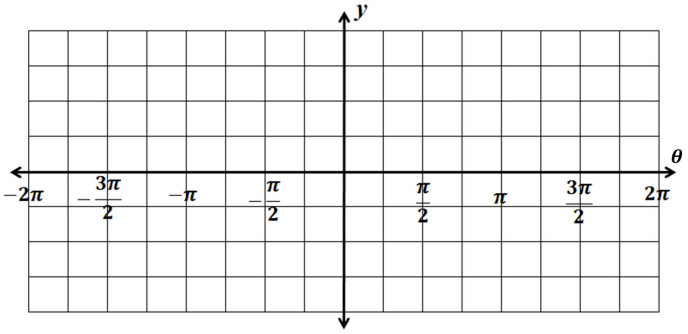
11. $y = -\frac{1}{2}\tan\theta$



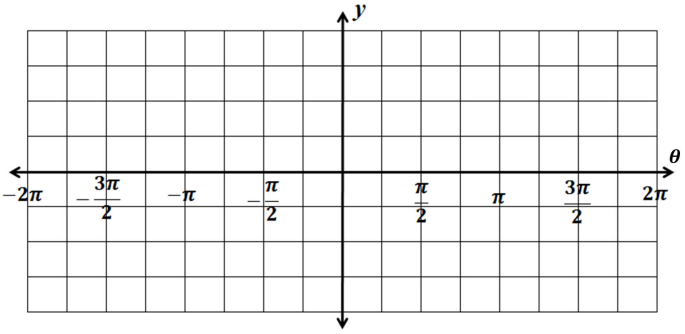
12. $y = 4\tan\left(\frac{1}{2}\theta\right) - 3$



13. $y = 2\tan\left(\theta - \frac{\pi}{4}\right) - 1$



14. $y = \tan(2\theta + \pi) + 2$



3.8 The Tangent Function

3.8 Test Prep

15. The graph of the function g is given in the xy-plane. If $g(x) = a\tan(bx) - 20$, where a and b are constants, which of the following could be true?

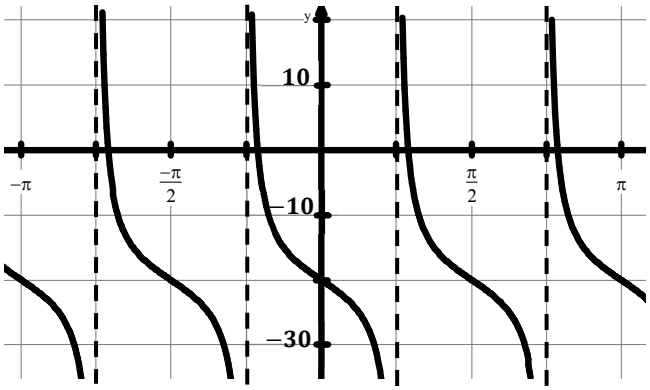
I. If $a > 0$ then $b > 1$
II. If $a > 0$ then $b < 0$
III. If $a < 0$ then $b > 1$
IV. If $a < 0$ then $b < 0$

(A) III only

(B) IV only

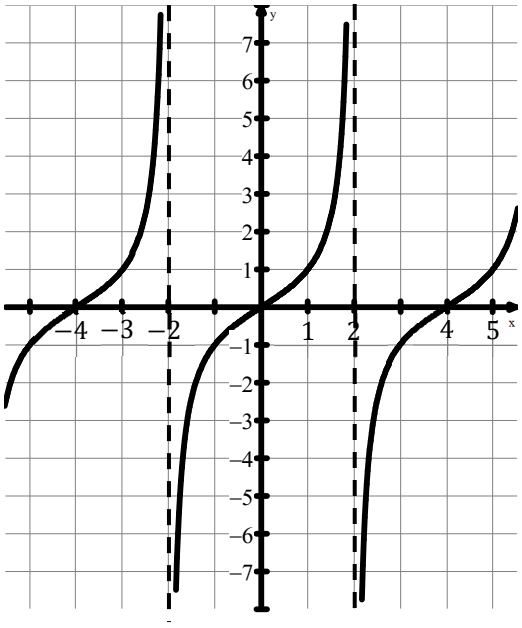
(C) I and IV only

(D) II and III only



Graph of g

16. The graph of $f(x) = \tan(bx)$, where b is a constant, is shown in the xy-plane. What is the value of b?



Graph of f

(A) 4

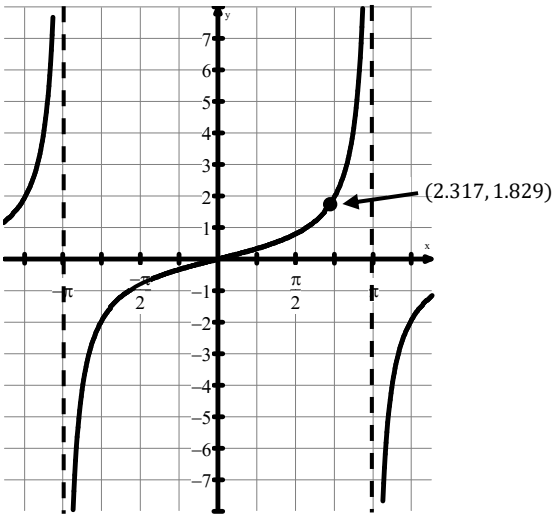
(B) 2

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4}$

---

17. The figures shows the graph of $f(x) = a\tan(bx)$, where a and b are constants, in the xy-plane. The graph of f has two vertical asymptotes at $x = -\pi$ and $x = \pi$, and a point with coordinates given is on the graph of f. What are all solutions to $f(x) = 1.829$?



Graph of f

(A) $x = 2.317$ only

(B) $x = 2.317 + \pi k$, where k is any integer

(C) $x = 2.317 + 2\pi k$, where k is any integer

(D) $x = 2.317 + 3\pi k$, where k is any integer

3.8 The Tangent Function

AP Precalculus Name: _____ **CA #1**

Write an equation that represents all asymptotes of the graph of f in the xy-plane.

1. $f(\theta) = \tan(8\theta)$

2. $f(\theta) = \tan\left(\frac{\theta}{2}\right)$

In the xy-plane, the angle θ is in standard position. What is the slope of the terminal ray of the angle?

3. $\theta = \frac{2\pi}{3}$

4. $\theta = \frac{5\pi}{4}$

Evaluate.

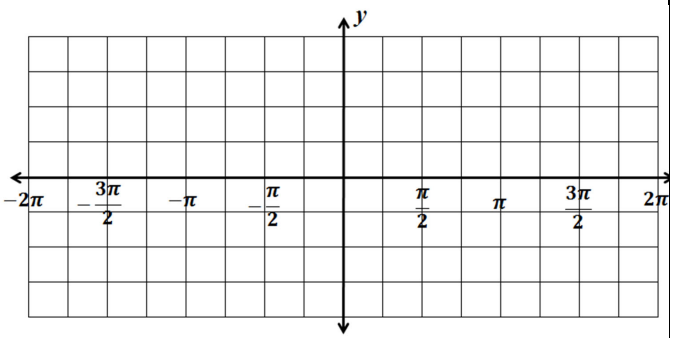
5. $\tan\frac{5\pi}{6}$

6. $\tan 2\pi$

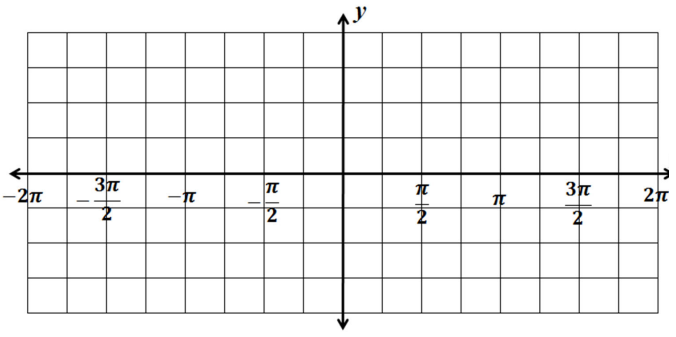
7. $\tan\frac{\pi}{6}$

Graph each trig function.

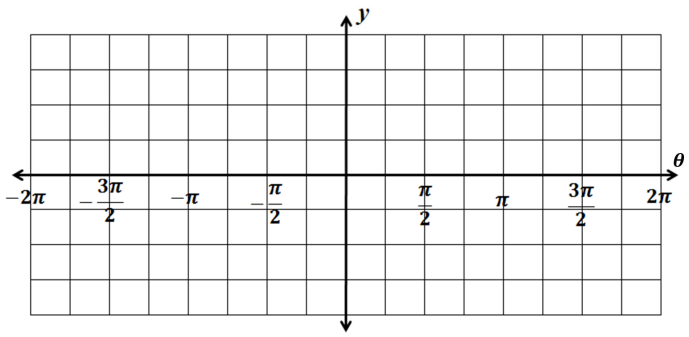
8. $y = \tan\left(\theta + \frac{\pi}{2}\right) - 2$



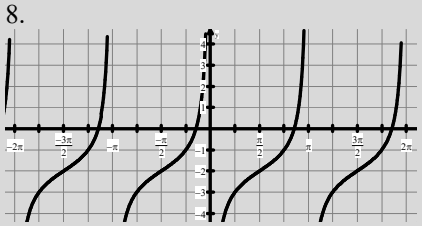
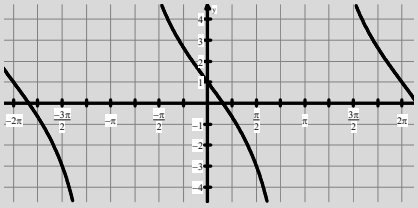
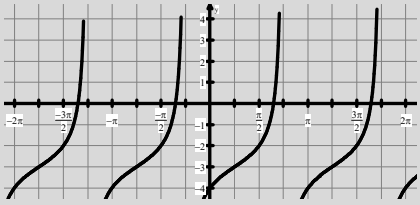
9. $y = 1 - 4\tan\left(\frac{\theta}{2}\right)$



10. $y = \tan\left(\theta - \frac{\pi}{4}\right) - 3$



Answers to 3.8 CA #1

1. $\theta = \frac{\pi}{16} + k\frac{\pi}{8}$, for integer values of k.	2. $\theta = \pi + k2\pi$, for integer values of k.	3. $-\sqrt{3}$	4. 1
5. $\frac{1}{-\sqrt{3}} = -\frac{\sqrt{3}}{3}$	6. 0	7. $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	
8. 	9. 	10. 	

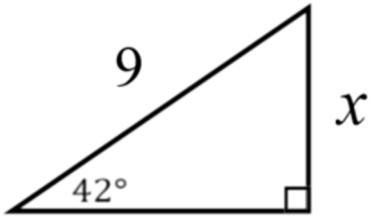
AP Precalc

Write your questions and thoughts here!

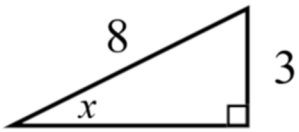
3.9 Inverse Trigonometric Functions

3.9 Notes

Find the missing side.



Find the missing angle.



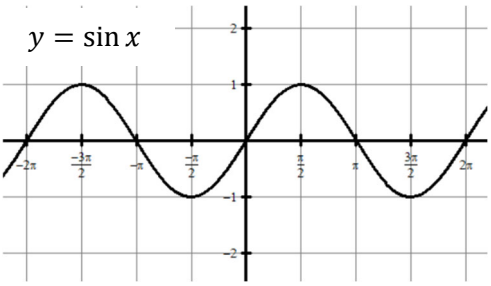
Trig Function

$f(x) = \sin(x)$

Domain:

Range:

Graph:



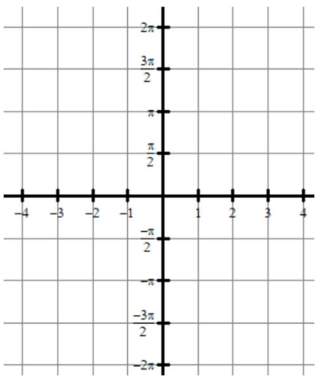
Inverse Trig Function

$f(x) = \sin^{-1}(x)$

Domain:

Range:

Graph:

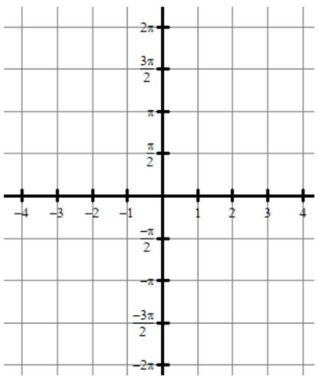


PRINCIPAL VALUES

Domain:

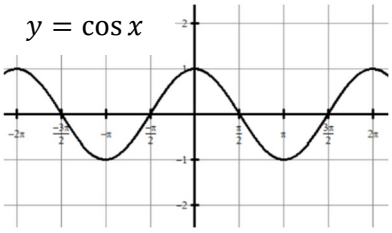
Range:

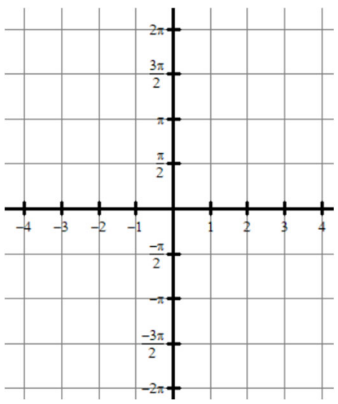
Graph:



$f(x) = \arccos(x)$

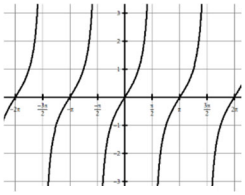
Restricted Domain:



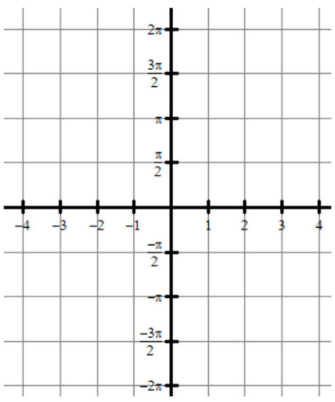


$f(x) = \arctan(x)$

Restricted Domain:



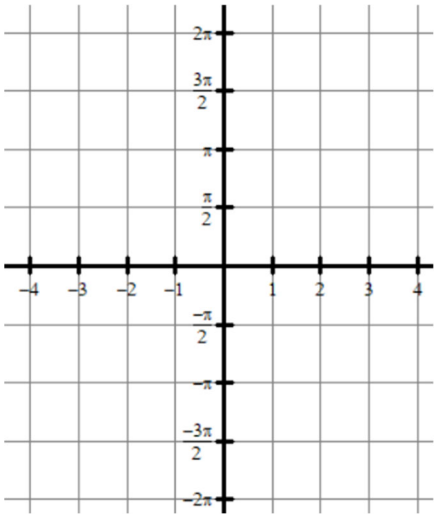
$y = \tan x$



Write your questions and thoughts here!

Find the inverse of the function. Graph it! State the domain and range of $f^{-1}(x)$.

$f(x) = 2\sin x + 1$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$



Evaluate the following expressions. Find the principal value in radians.

$\sin^{-1}\left(\frac{1}{2}\right)$	$\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$	$\tan^{-1}\left(\sqrt{3}\right)$

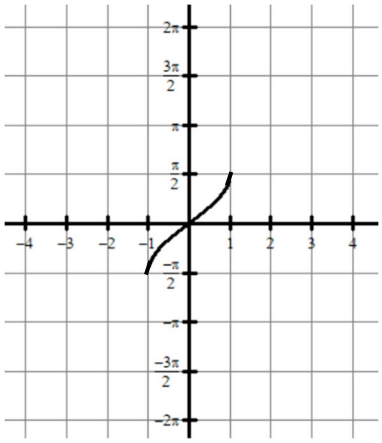
3.9 Inverse Trigonometric Functions

AP Precalculus

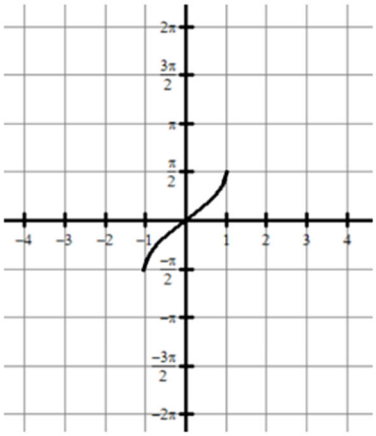
3.9 Practice

The graph of $f(x) = \sin^{-1}(x)$ is shown below. Use the graph of f to graph $g(x)$.

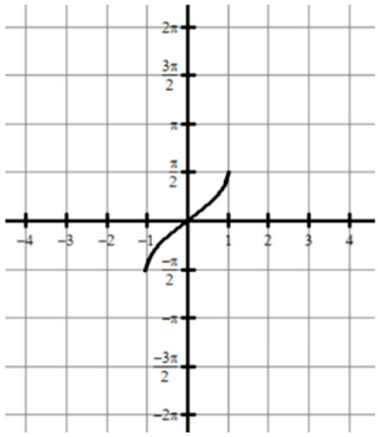
1. $g(x) = 2\sin^{-1}(x)$



2. $g(x) = 3\sin^{-1}(x) - \frac{\pi}{2}$

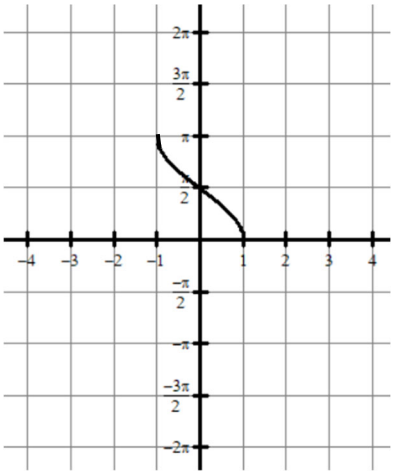


3. $g(x) = \sin^{-1}\left(\frac{1}{3}(x-1)\right)$

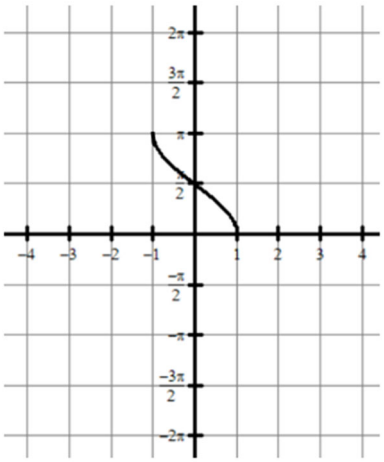


The graph of $f(x) = \cos^{-1}(x)$ is shown below. Use the graph of f to graph $g(x)$.

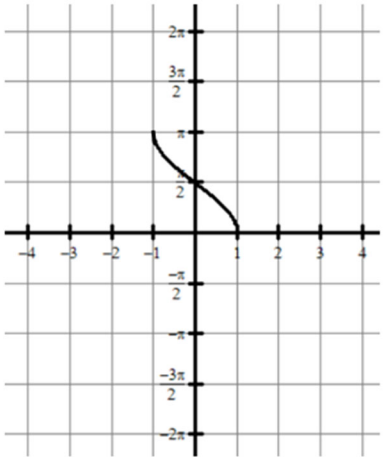
4. $g(x) = 2\cos^{-1}(x) - \pi$



5. $g(x) = \cos^{-1}(x+2) - \frac{\pi}{2}$

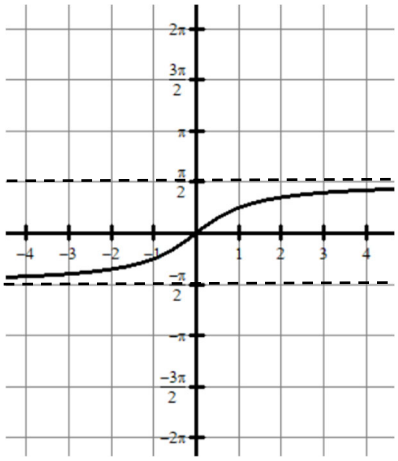


6. $g(x) = 2\cos^{-1}(-2x)$

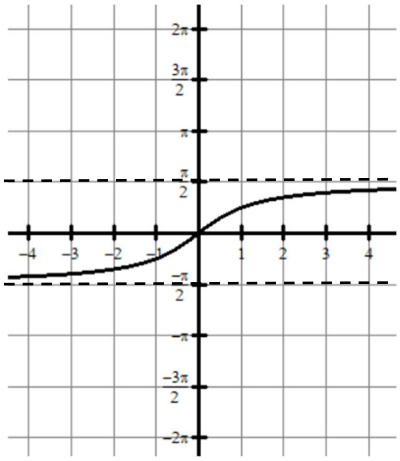


The graph of $f(x) = \tan^{-1}(x)$ is shown below. Use the graph of f to graph $g(x)$.

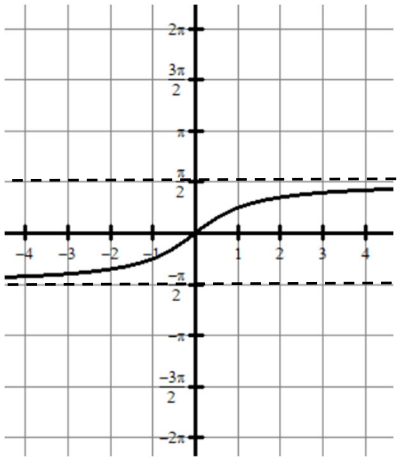
7. $g(x) = 3\tan^{-1}(x)$



8. $g(x) = 2\tan^{-1}(x) + \pi$



9. $g(x) = \frac{1}{2}\tan^{-1}(-x) - \pi$



Find the inverse of each function and list the domain and range of $f^{-1}(x)$.

10. $f(x) = 5\sin x - 3$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

11. $f(\theta) = \frac{1}{2}\tan\theta$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

12. $f(x) = 4 - \cos(2x)$ for $0 \le x \le \frac{\pi}{2}$

13. $f(\theta) = 3\sin(\theta - \pi) + 1$ for $\frac{\pi}{2} \le \theta \le \frac{3\pi}{2}$

Evaluate the following expressions. Find the principal value in radians. NO CALCULATOR!!

14. $\sin^{-1}\left(\frac{1}{2}\right)$

15. $\cos^{-1}\left(\frac{1}{2}\right)$

16. $\tan^{-1}(-1)$

17. $\arccos\left(\frac{\sqrt{3}}{2}\right)$

18. $\arcsin\left(-\frac{\sqrt{2}}{2}\right)$

19. $\tan^{-1}(-\sqrt{3})$

20. $\sin^{-1}(1)$

21. $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

22. $\arccos\left(-\frac{\sqrt{3}}{2}\right)$

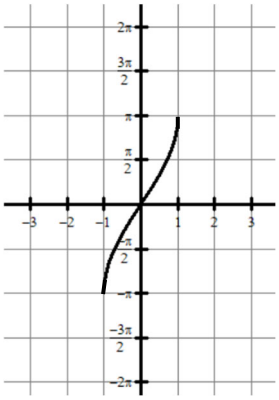
23. $\tan^{-1}\left(-\frac{\sqrt{3}}{3}\right)$

24. $\cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$

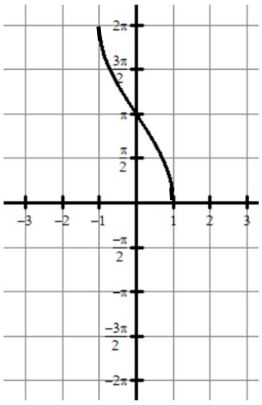
25. $\arcsin(0)$

26. Which of the following is the graph of $f(x) = 2\cos^{-1}(x)$?

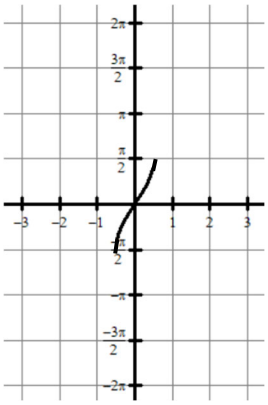
(A)



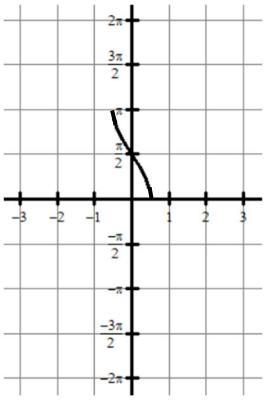
(B)



(C)

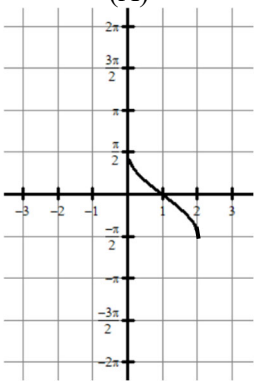


(D)

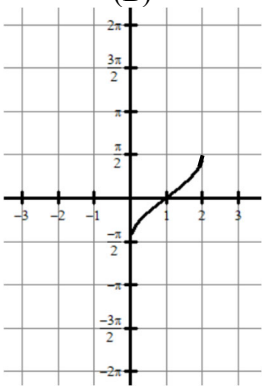


27. Given $f(x) = 1 - \sin(x)$, which of the following is the graph of $f^{-1}(x)$?

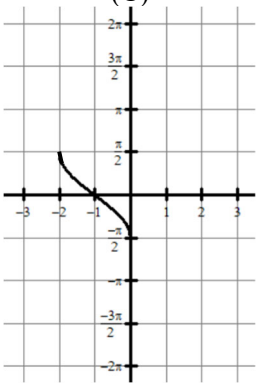
(A)



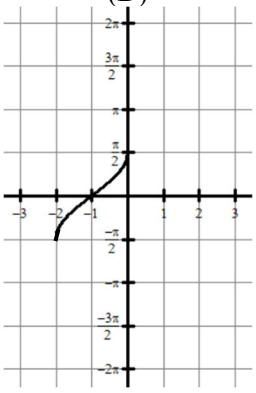
(B)



(C)



(D)



28. Given $f(x) = 2\sin\left(\frac{1}{3}x\right) + 1$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$, what is the domain of $f^{-1}(x)$?

(A) $\left[-\frac{3\pi}{2}, \frac{3\pi}{2}\right]$

(B) $[0, 2]$

(C) $[-1, 3]$

(D) $[-\pi, \pi]$

3.9 Inverse Trigonometric Functions

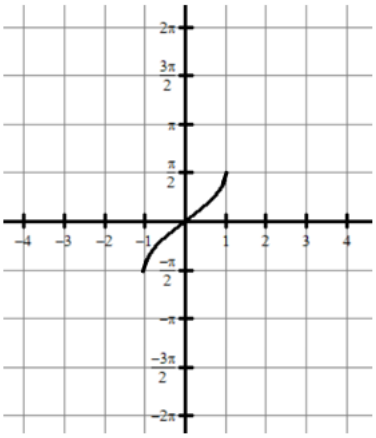
AP Precalculus

Name: _____

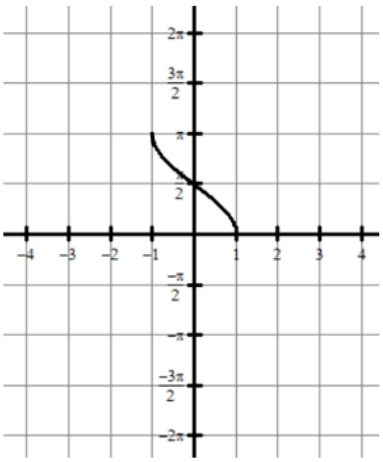
CA #1

The parent function is shown below. Use the parent function to graph $g(x)$.

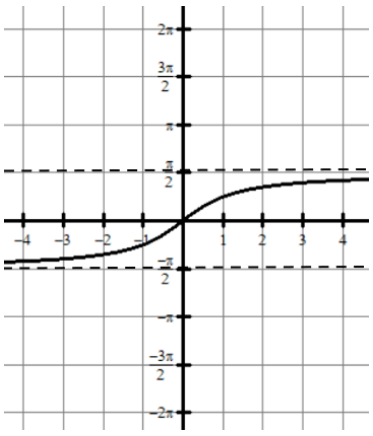
1. $g(x) = \sin^{-1}\left(\frac{1}{2}x\right) - \pi$



2. $g(x) = 2\cos^{-1}(x-1)$



3. $g(x) = 3\tan^{-1}(-x)$



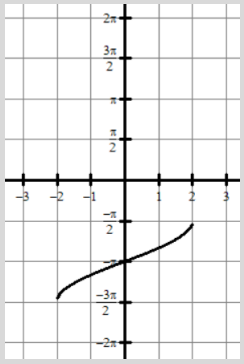
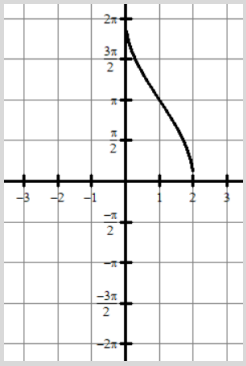
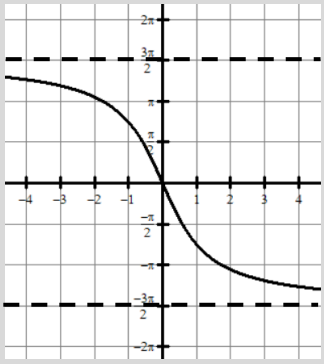
Find the inverse of each function and list the domain and range of $f^{-1}(x)$.

4. $f(x) = 2\sin x + 1$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

5. $f(\theta) = \frac{1}{3}\cos(2\theta)$ for $0 \le \theta \le \frac{\pi}{2}$

Evaluate the following expressions. Find the principal value in radians. NO CALCULATOR!!		
6. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$	7. $\cos^{-1}(0)$	8. $\arctan(\sqrt{3})$
9. $\arccos\left(\frac{\sqrt{2}}{2}\right)$	10. $\sin^{-1}\left(-\frac{1}{2}\right)$	11. $\arccos(-1)$

Answers to 3.9 CA #1

1. 	2. 	3. 	4. $f^{-1}(x) = \sin^{-1}\left(\frac{x-1}{2}\right)$ Domain: $-1 \le x \le 3$ Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$
5. $f^{-1}(\theta) = \frac{1}{2}\cos^{-1}(3\theta)$ Domain: $-\frac{1}{3} \le x \le \frac{1}{3}$ Range: $0 \le y \le \frac{\pi}{2}$	6. $-\frac{\pi}{3}$	7. $\frac{\pi}{2}$	8. $\frac{\pi}{3}$
9. $\frac{\pi}{4}$	10. $-\frac{\pi}{6}$	11. π	

3.10 Trig Equations and Inequalities

3.10 Notes

Write your questions and thoughts here!

Solve Trigonometric Equations

RESTRICTED DOMAIN

$$0 \leq x \leq 2\pi$$

Exact Values

$2\sin x + 5 = 4$

Approximate Values

$\tan^2 x + 2\tan x - 8 = 0$

ALL VALUES

Exact Values

$4\cos^2 \theta = 2$

Approximate Values

$3\sin^2 \theta = \sin \theta$

Write your questions and thoughts here!

Solve Trigonometric Inequalities

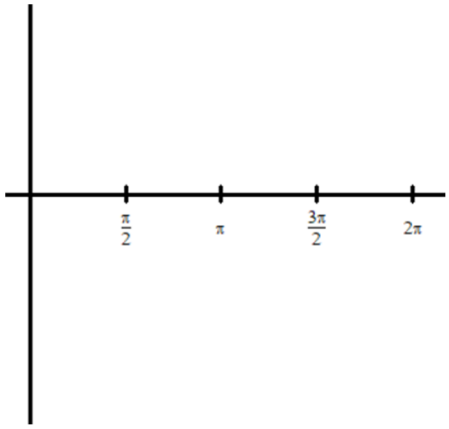
RESTRICTED DOMAIN

$$0 \leq x \leq 2\pi$$

Exact Values

$4\cos x - 1 \leq -3$

Include a rough sketch.



CHANGE IN PERIOD

ALL EXACT VALUES

$\cos(2\theta) + 4 = 4$

3.10 Trigonometric Equations and Inequalities

AP Precalculus

3.10 Practice

Solve each equation for $0 \leq x \leq 2\pi$. Find the exact value(s) using the unit circle.

1. $2\sin x + 3 = 4$

2. $4 - 3\cos x = 7$

3. $\tan^2 x = 1$

4. $-1 = \cos^2 x + 2\cos x$

Solve each equation for $0 \leq \theta \leq 2\pi$. Find the approximate value(s) using a calculator.

5. $6 = 3\cos\theta + 7$

6. $5\sin^2\theta + 3 = 6$

7. $6\cos^2\theta + 4\cos\theta = 0$

8. $\cos\theta = 3\cos^2\theta$

Solve each equation. Find ALL exact value(s) using the unit circle.

9. $2\sin^2\theta = \sin\theta$

10. $4\tan^2(2x) = 12$

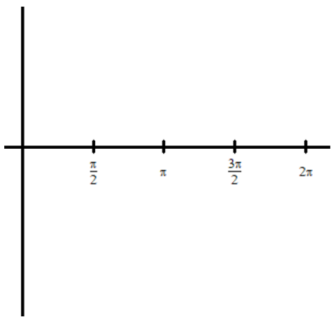
Solve each equation. Find ALL approximate value(s) using a calculator.

11. $7\sin^2 x = 5$

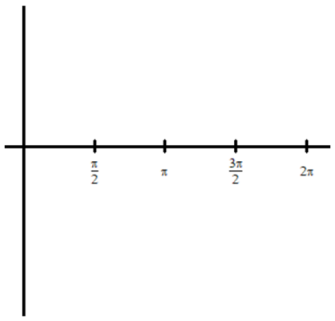
12. $5\sin\theta + 3 = 4$

Solve each inequality for $0 \le x \le 2\pi$. Find the exact value(s). Include a rough sketch.

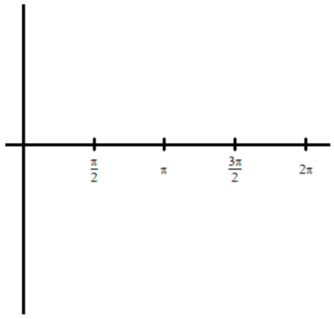
13. $2\sin x + 2 > 3$



14. $2\cos x \ge \sqrt{3}$



15. $2\cos\left(x - \frac{\pi}{2}\right) - 2 \le -3$



16. What are all values of θ, for $0 \le \theta \le 2\pi$, where $4\sin^2\theta = 2\sin\theta$?

(A) $0, \frac{\pi}{3}, \pi, \frac{2\pi}{3}, 2\pi$

(B) $0, \frac{\pi}{6}, \pi, \frac{5\pi}{6}, 2\pi$

(C) $\frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{2}$

(D) $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

17. The function f is given by $f(x) = 2\cos(3x) - 1$. For how many values of x where $0 \le x \le 2\pi$ does $f(x) = 0$?

(A) None

(B) Two

(C) Four

(D) Six

18. The function g is given by $g(x) = 2\cos(x)$. What are all solutions $g(x) = \sqrt{3}$?

(A) $x = \frac{\pi}{6} + 2\pi k$ and $\frac{5\pi}{6} + 2\pi k$, where k is any integer

(B) $x = \pm\frac{\pi}{6} + 2\pi k$, where k is any integer

(C) $x = \frac{\pi}{3} + 2\pi k$ and $\frac{2\pi}{3} + 2\pi k$, where k is any integer

(D) $x = \pm\frac{\pi}{3} + 2\pi k$, where k is any integer

19. What is the solution set for $4 < 2\sin(x) + 3$ where $0 \le x \le 2\pi$?

(A) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$

(B) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$

(C) $\left(\frac{\pi}{3}, \frac{2\pi}{3}\right)$

(D) $\left(0, \frac{\pi}{3}\right) \cup \left(\frac{2\pi}{3}, 2\pi\right)$

20. What are all values of θ, for $0 \le \theta \le 2\pi$, where $\sin^2\theta - \cos\theta\sin\theta = 0$?

(A) $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

(B) $0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}, 2\pi$

(C) $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$

(D) $\frac{\pi}{4}$ and $\frac{5\pi}{4}$

3.10 Trigonometric Equations and Inequalities

AP Precalculus Name: _____

CA #1

Solve each equation for $0 \le \theta \le 2\pi$. Find the exact value(s) using the unit circle

1. $2\sin\theta - 3 = -2$

2. $\cos^2\theta + \cos\theta = 0$

Solve each equation for $0 \le x \le 2\pi$. Find the approximate value(s) using a calculator.

3. $5 + 3\sin x = 7$

4. $\tan^2 x - 3\tan x = 10$

Solve each equation. Find ALL exact value(s) using the unit circle

5. $3 = 1 - 2\cos(2x)$

6. $4\sin^2\theta - 1 = 2$

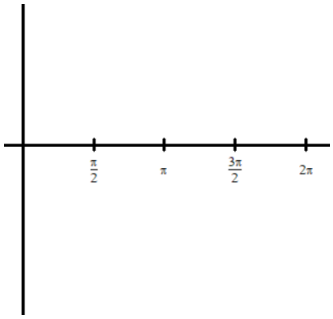
Solve each equation. Find ALL approximate value(s) using a calculator.

7. $5 - 4\cos\theta = 6$

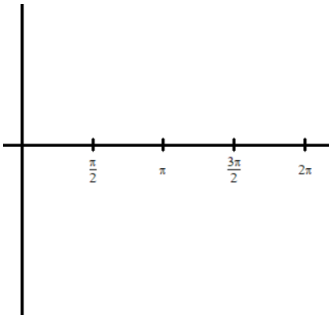
8. $\tan^2 x = 3\tan x$

Solve each inequality for $0 \le x \le 2\pi$. Find the exact value(s). Include a rough sketch.

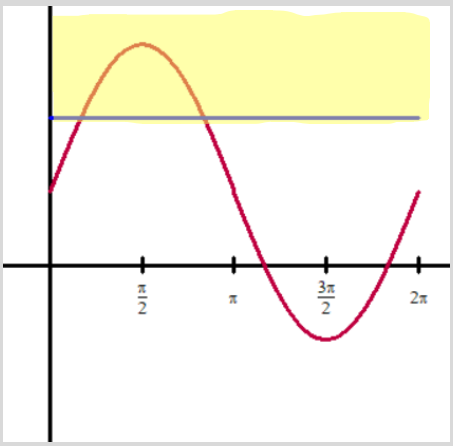
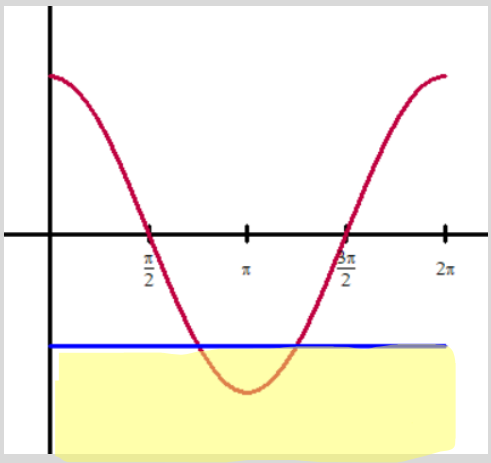
9. $2\sin x + 1 > 2$



10. $2\cos x \le -\sqrt{2}$



Answers to 3.10 CA #1

1. $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$	2. $\theta = \frac{\pi}{2}, \theta = \frac{3\pi}{2}$ and $\theta = \pi$	3. $x = 0.729$ and $x = 2.412$
4. $x = 1.373$ and $x = 4.514$ $x = 2.034$ and $x = 5.176$	5. $x = \frac{\pi+2\pi n}{2} = \frac{\pi}{2} + \pi n$ where n is an integer	6. $x = \frac{\pi}{3} + \pi n$ $x = \frac{2\pi}{3} + \pi n$ where n is an integer
7. $\theta = 1.823 + 2\pi n$ $\theta = 4.460 + 2\pi n$ where n is an integer		8. $\theta = 0 + \pi n$ $\theta = 1.249 + \pi n$ where n is an integer
9. $\frac{\pi}{6} < x < \frac{5\pi}{6}$ 		10. $\frac{3\pi}{4} \le x \le \frac{5\pi}{4}$ 

AP Precalc

Write your questions and thoughts here!

3.11 Secant, Cosecant, Cotangent Functions

3.11 Notes

RECIPROCAL TRIG FUNCTIONS

Cosecant	Secant	Cotangent
$\csc x =$	$\sec x =$	$\cot x =$

Find the exact value of the following expressions.

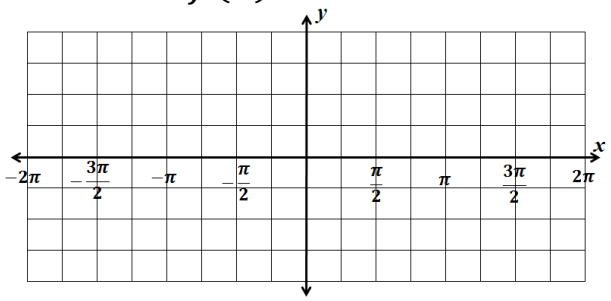
$\csc\left(\frac{\pi}{6}\right)$	$\sec\left(\frac{3\pi}{4}\right)$	$\cot\left(\frac{4\pi}{3}\right)$

Find the approximate value of the following expressions.

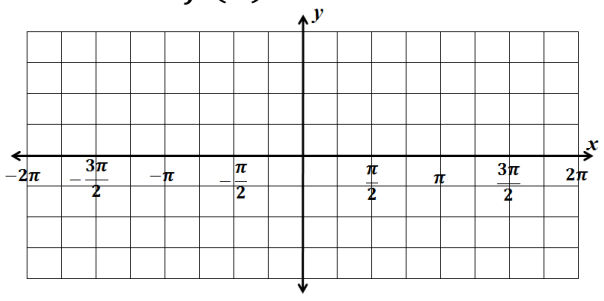
$\csc\left(\frac{\pi}{5}\right)$	$\sec(-1.43)$	$\cot(\pi)$

Graphs of reciprocal trig functions.

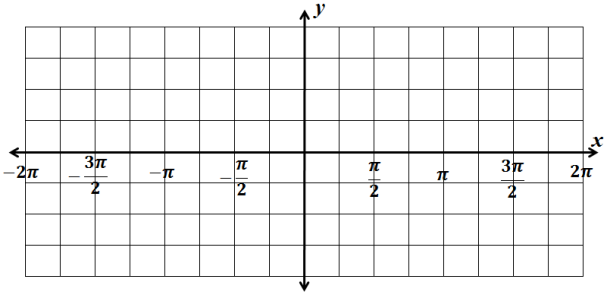
$f(x) = \csc x$



$f(x) = \sec x$



$f(x) = \cot x$



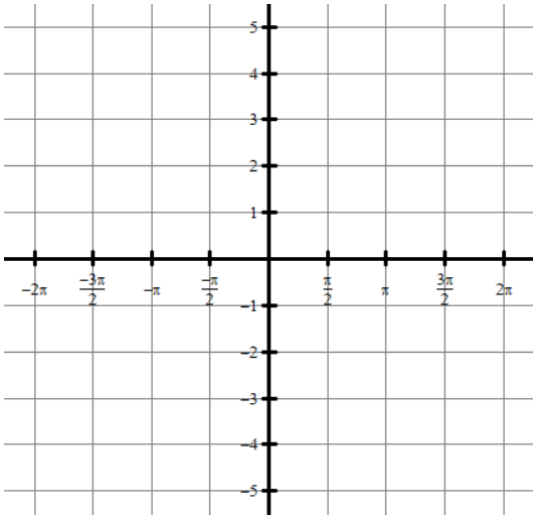
Write your questions and thoughts here!

Graph the following. State the range and all vertical asymptotes.

$$f(x) = 3\sec x - 1$$

Range:

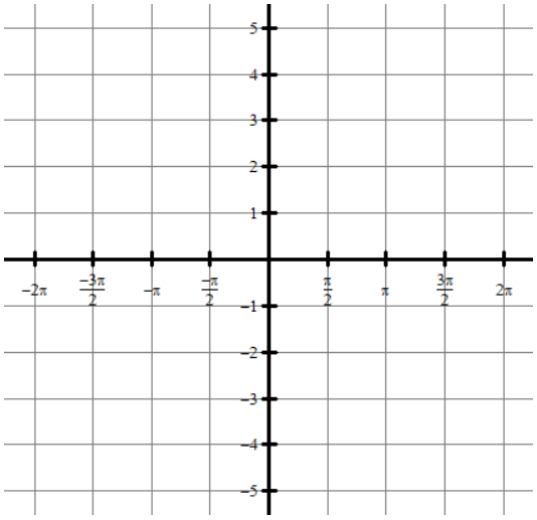
Vertical Asymptotes:



$$f(x) = 2\csc\left(x - \frac{\pi}{2}\right) + 1$$

Range:

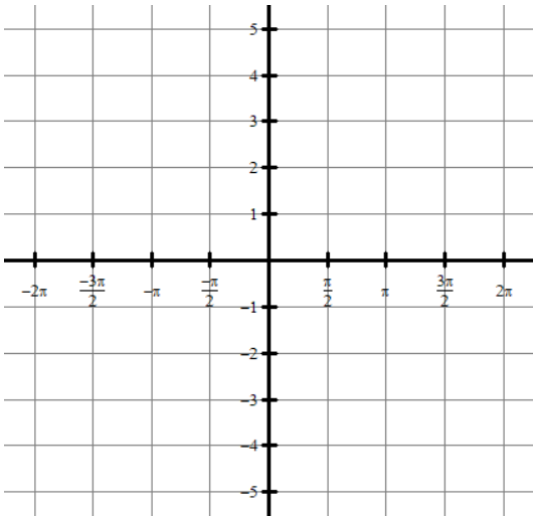
Vertical Asymptotes:



$$f(x) = 2\cot\big(2(x + \pi)\big)$$

Range:

Vertical Asymptotes:



3.11 Secant, Cosecant, and Cotangent Functions

AP Precalculus

3.11 Practice

Evaluate the following expressions. Use exact values.

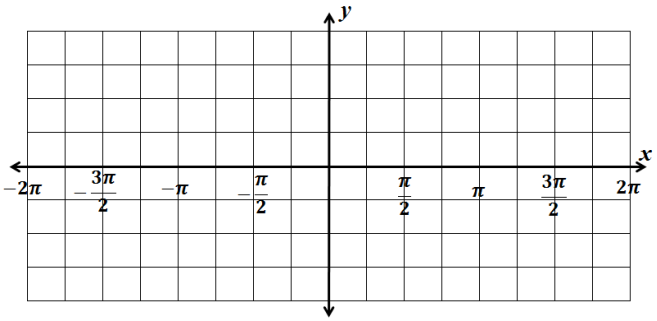
1. $\sec\left(\frac{\pi}{3}\right)$	2. $\csc\left(\frac{\pi}{4}\right)$	3. $\cot\left(\frac{3\pi}{2}\right)$
4. $\sec\left(\frac{7\pi}{6}\right)$	5. $\csc\left(-\frac{2\pi}{3}\right)$	6. $\cot\left(\frac{3\pi}{4}\right)$
7. $\csc(\pi)$	8. $\sec\left(\frac{5\pi}{4}\right)$	9. $\cot\left(\frac{5\pi}{3}\right)$

Evaluate the following expressions. Use approximate values from calculator.

10. $\csc(1.43)$	11. $\sec\left(\frac{\pi}{5}\right)$	12. $\cot\left(\frac{5\pi}{7}\right)$
13. $\sec(1.3\pi)$	14. $\cot(-3.26)$	15. $\csc\left(\frac{\pi}{9}\right)$

Graph the following and state all vertical asymptotes.

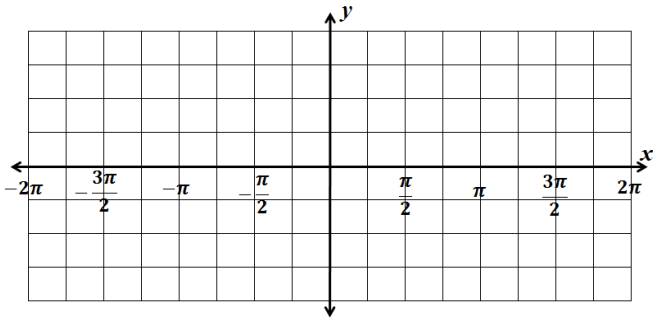
16. $f(x) = 2\csc x$



Range:

Vertical Asymptotes:

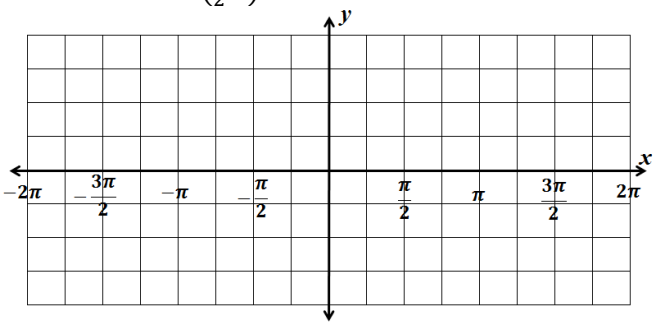
17. $f(x) = \sec(x) - 1$



Range:

Vertical Asymptotes:

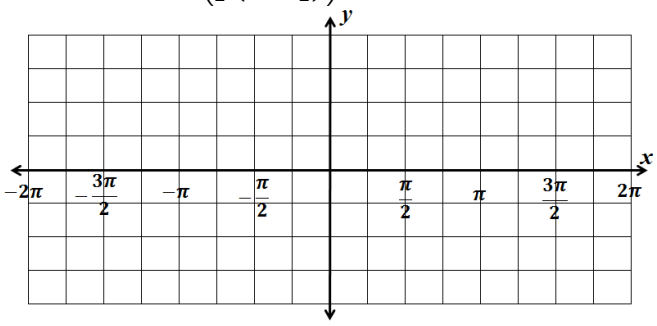
18. $f(x) = \cot\left(\frac{1}{2}x\right)$



Range:

Vertical Asymptotes:

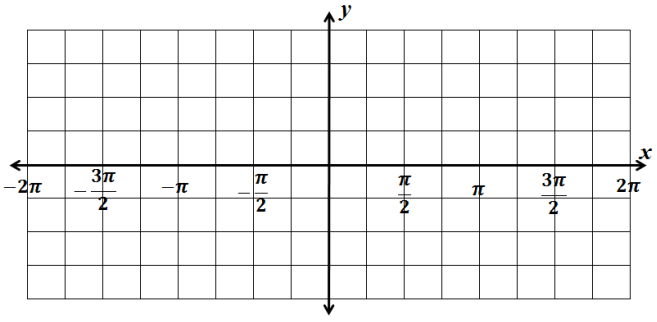
19. $f(x) = 2\csc\left(\frac{1}{2}\left(x - \frac{\pi}{2}\right)\right)$



Range:

Vertical Asymptotes:

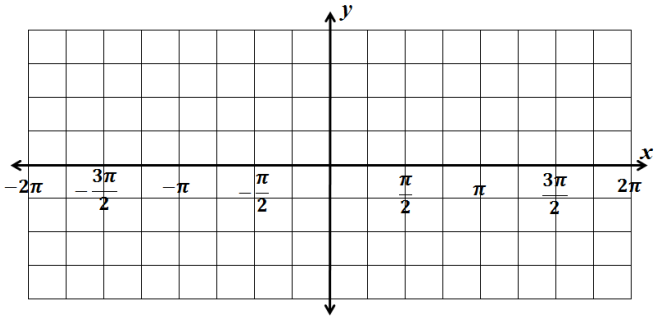
20. $f(x) = 2\cot(x) - 1$



Range:

Vertical Asymptotes:

21. $f(x) = \sec(2x) + 1$

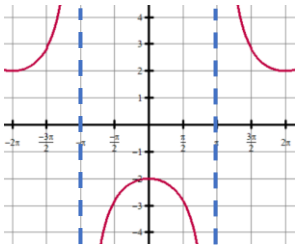


Range:

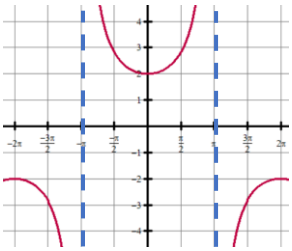
Vertical Asymptotes:

21. Which of the following is the graph of $f(x) = -2\csc\left(\frac{1}{2}x\right)$?

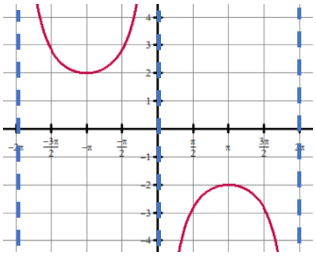
(A)



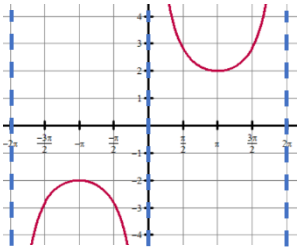
(B)



(C)

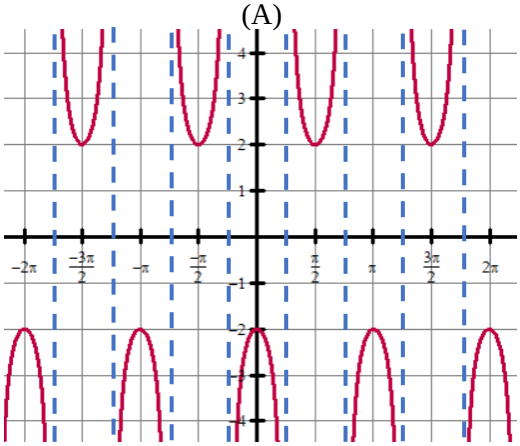


(D)

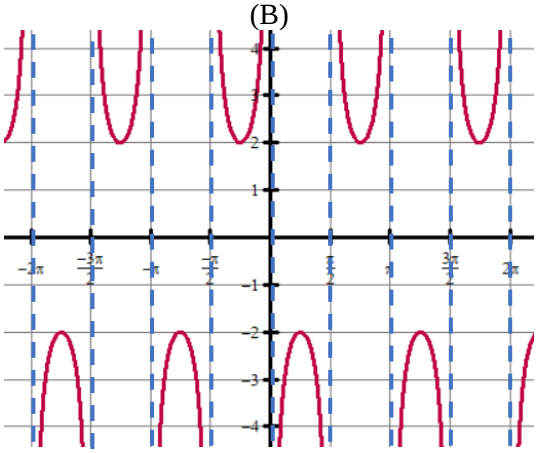


22. Which of the following is the graph of $f(\theta) = 2\sec\left(2\left(\theta + \frac{\pi}{2}\right)\right)$?

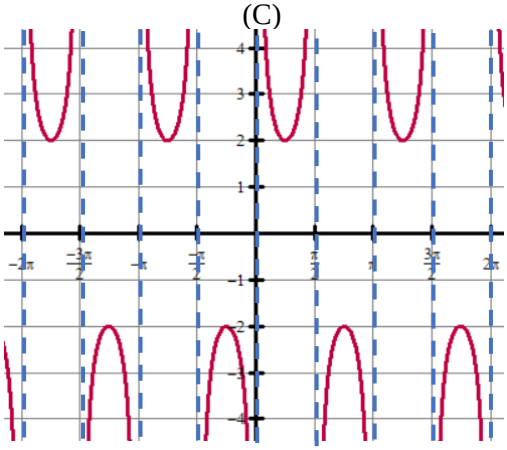
(A)



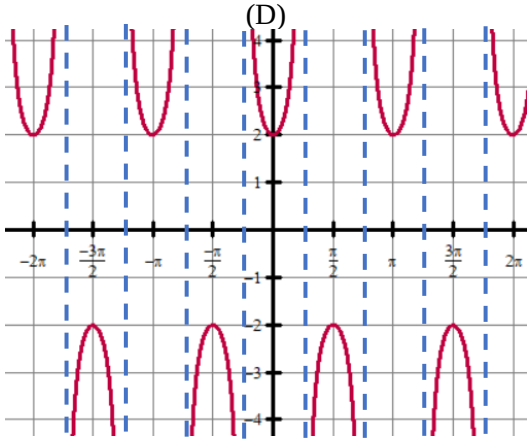
(B)



(C)



(D)



23. Which of the following describes the graph of $f(x) = 2\cot x$?

(A) Vertical asymptotes at $x = \frac{\pi}{2} + \pi k$, where k is an integer, and the range is all real numbers.

(B) Vertical asymptotes at $x = \frac{\pi}{2} + \pi k$, where k is an integer, and the range is $(-\infty, -2] \cup [2, \infty)$.

(C) Vertical asymptotes at $x = \pi + \pi k$, where k is an integer, and the range is all real numbers.

(D) Vertical asymptotes at $x = \pi + \pi k$, where k is an integer, and the range is $(-\infty, -2] \cup [2, \infty)$.

3.11 Secant, Cosecant, Cotangent Functions

AP Precalculus Name: _____ **CA #1**

Evaluate the following expressions. Use exact values.

1. $\sec\left(\frac{4\pi}{3}\right)$

2. $\csc\left(\frac{3\pi}{2}\right)$

3. $\cot\left(\frac{\pi}{3}\right)$

Evaluate the following expressions. Use approximate values from calculator.

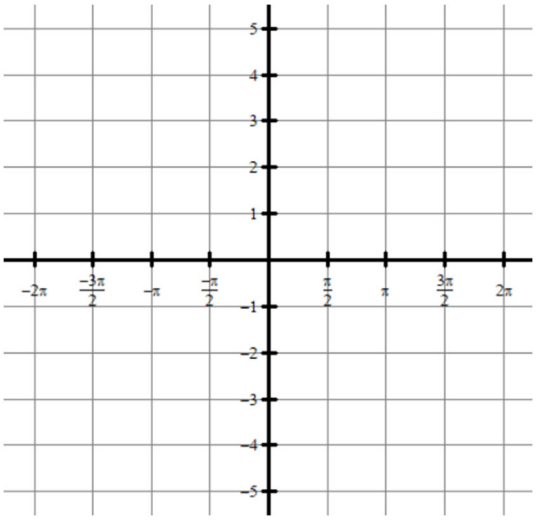
4. $\csc(1.79)$

5. $\sec\left(\frac{2\pi}{5}\right)$

6. $\cot\left(\frac{4\pi}{7}\right)$

Graph the following. State the range and all vertical asymptotes.

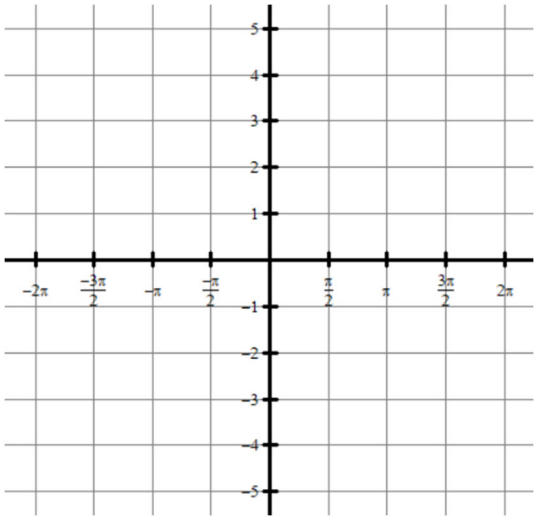
7. $f(x) = 2\csc(x) - 1$



Range:

Vertical Asymptotes:

8. $f(x) = 2\sec\left(\frac{1}{2}x\right)$



Range:

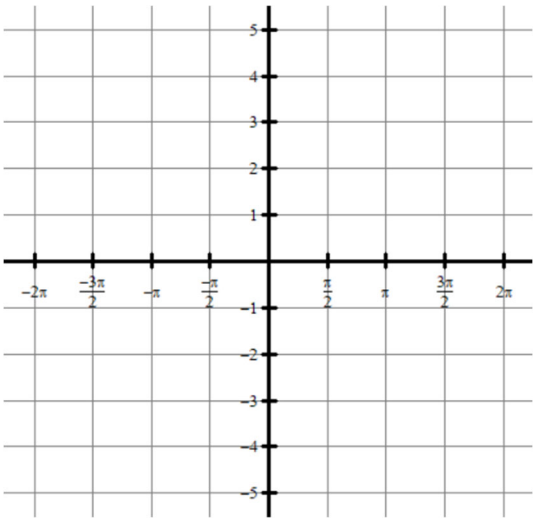
Vertical Asymptotes:

Graph the following. State the range and all vertical asymptotes.

9. $f(x) = \cot(x - \pi)$

Range:

Vertical Asymptotes:



Answers to 3.11 CA #1

1. -2	2. -1	3. $\frac{\sqrt{3}}{3}$
4. 1.024	5. 3.236	6. -0.228

7.

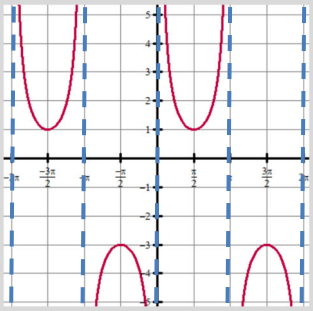
Range:

$(-\infty, -3] \cup [1, \infty)$

Vertical:

$x = \pi n$

where n is an integer



8.

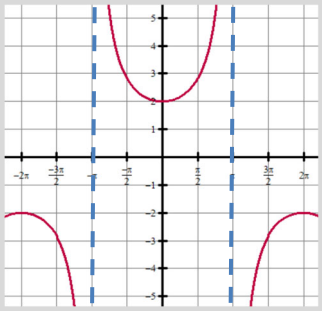
Range:

$(-\infty, -2] \cup [2, \infty)$

Vertical:

$x = \pi + 2\pi n$

where n is an integer



9.

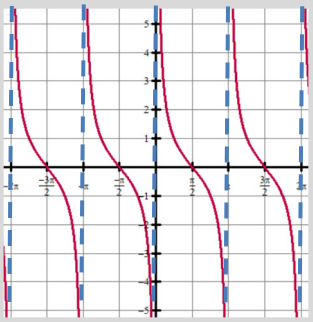
Range:

$(-\infty, \infty)$

Vertical:

$x = \pi n$

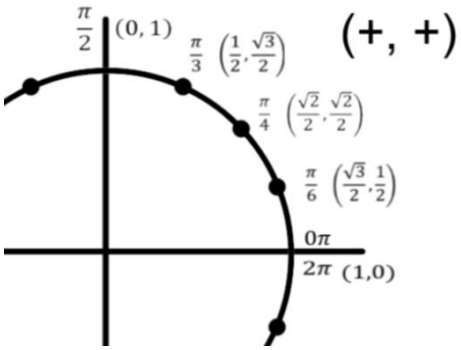
where n is an integer



3.12A Equivalent Representations of Trig Functions 3.12A Notes

Pythagorean Identities

Unit circle



Reciprocal Pythagorean Identities

$\sin^2 x + \cos^2 x = 1$

$\sin^2 x + \cos^2 x = 1$

Use trig identities to write each expression in terms of a single trig identity.

$\tan x \cos x$

$\frac{1 - \cos^2 x}{\cos^2 x}$

$\frac{\cos^2 x + \sin^2 x}{\sin x \sec x}$

Tips:

1. **Rewrite in $\sin x$ and $\cos x$**
2. **Squared trig functions could be Pythagorean Identity**
3. **Put in terms of one trig function**

Write your questions and thoughts here!

Use trig identities to solve the trig equations for $0 \le x \le 2\pi$. Find exact values.

$1 = \sec^2 x + \tan x$

$1 - \sin^2 x + \cos^2 x = 2$

3.12A Equivalent Representations of Trig Functions

AP Precalculus

3.12A Practice

Use trig identities to write each expression in terms of a single trig identity.

1. $\cot x \sin x$
2. $\frac{1-\sin^2 x}{\sin^2 x}$
3. $\sin^2 x + \cos^2 x + \tan^2 x$
4. $\sin\theta \sec\theta$
5. $\frac{\sec^2 x - 1}{\sin^2 x}$
6. $\cot^2\theta\,(1 - \cos^2\theta)$

Use trig identities to solve the trig equations for $0 \leq x \leq 2\pi$. Find exact values.

7. $\sin^2 x - \cos x = 1$

8. $2 \cos x \tan x = 1$

9. $\sec^2 x + \tan^2 x = 1$

10. $\sin^3 x \, (1 + \cot^2 x) = 1$

11. $\frac{\cos x}{\sec x} = \frac{3}{4}$

12. $(1 - \sin^2 x) \sec x = \frac{\sqrt{2}}{2}$

13. The function f is given by $f(x) = \frac{\csc^2 x}{\sec^2 x}$. Which of the following expressions is equivalent to $f(x)$?

(A) $\cos^2 x$

(B) $\tan^2 x$

(C) $\sin^2 x$

(D) $\cot^2 x$

14. The function f is given by $f(\theta) = \frac{\csc\theta \cot\theta \sin\theta}{\cos\theta}$. Which of the following expressions is equivalent to $f(\theta)$?

(A) $\csc\theta$

(B) $\sin\theta$

(C) $\sec\theta$

(D) $\tan\theta$

15. The function g is defined by $g(x) = \csc^2 x + \cot x$. What are all solutions to $g(x) = 1$ on the interval $0 \leq x \leq 2\pi$?

(A) $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$

(B) $x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{4}$

(C) $x = \frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$

(D) $x = \frac{\pi}{2}$

16. The function g is given by $g(x) = \frac{1-\cos^2 x}{\cos^2 x} + 1$. Which of the following expressions is equivalent to $g(x)$?

(A) $\cot^2 x$

(B) $\tan^2 x$

(C) $\csc^2 x$

(D) $\sec^2 x$

3.12A Equivalent Representations of Trig Functions

AP Precalculus Name: _____ **CA #1**

Use trig identities to write each expression in terms of a single trig identity.

1. $\sin x \sec x$

2. $(1 - \sin^2 x) \sec x$

3. $\sin^2 x + \cos^2 x + \cot^2 x$

4. $\frac{\sin\theta \csc\theta}{\cot\theta}$

5. $\frac{\sec^2 x - 1}{\cot x}$

6. $\frac{\sec\theta}{\cos\theta}$

Use trig identities to solve the trig equations for $0 \leq x \leq 2\pi$. Find exact values.

7. $\sec^2 x - 1 = \tan x$

8. $2\cos^2 x \sec x = 1$